



EUROPEAN
SPALLATION
SOURCE

Guides and gravity in McStas

Peter Willendrup



Adapted from slides by Mads Bertelsen, ESS



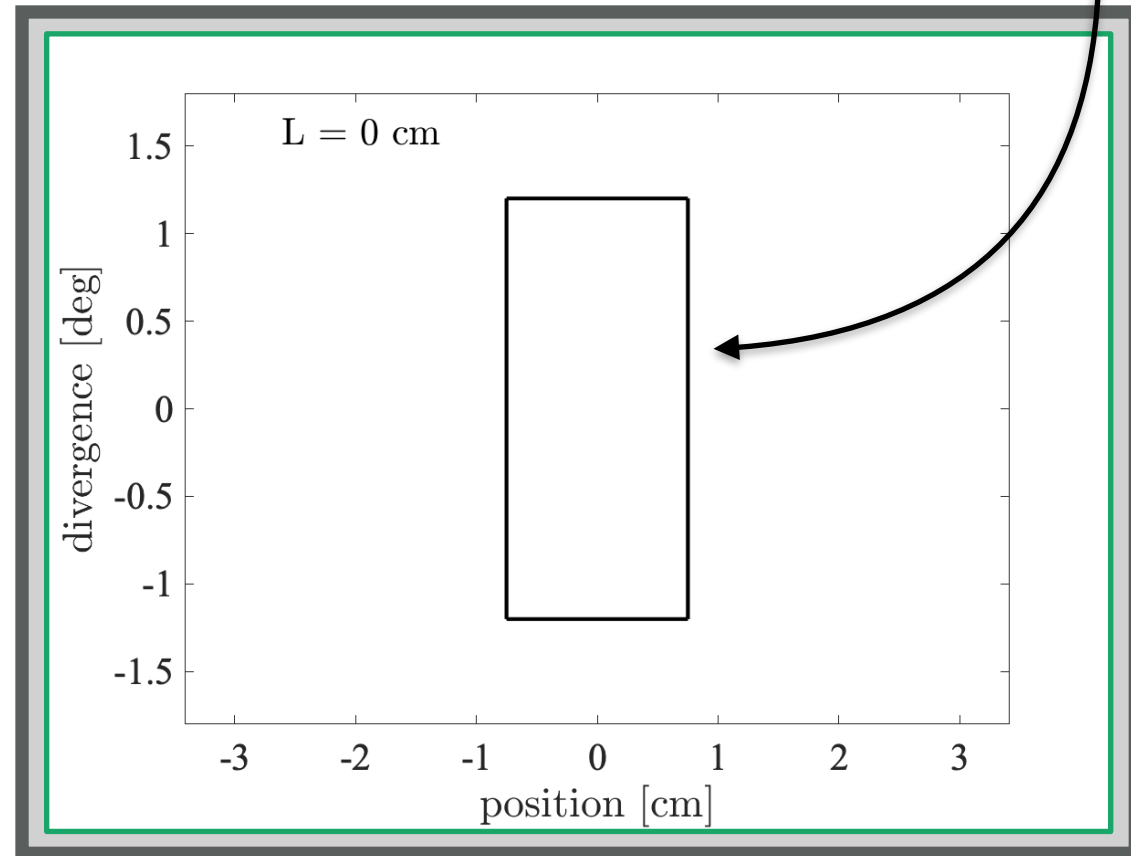
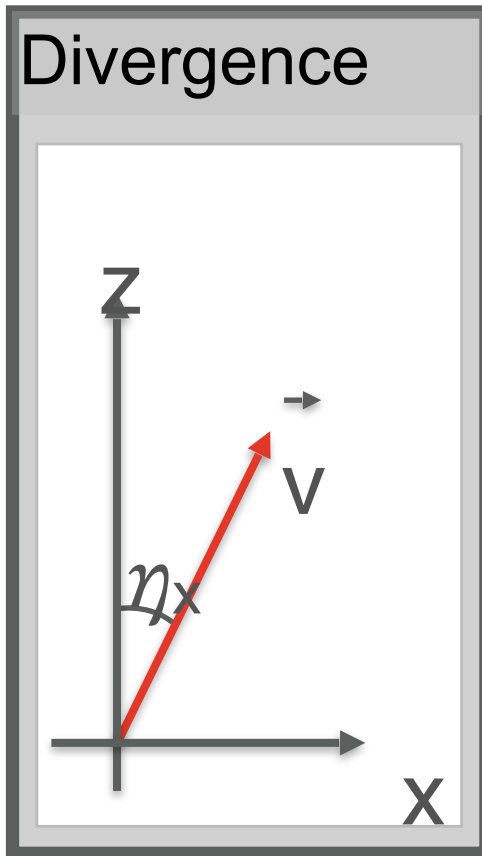
Overview

- Description of phase-space and propagation
- Reflectivity
- McStas coordinate system
- Gravitation in McStas
- Guide components with support for gravity
 - Guide_gravity
 - Elliptic_guide_gravity
- Breaking line of sight
- Example
- Exercise

**2025 ISIS
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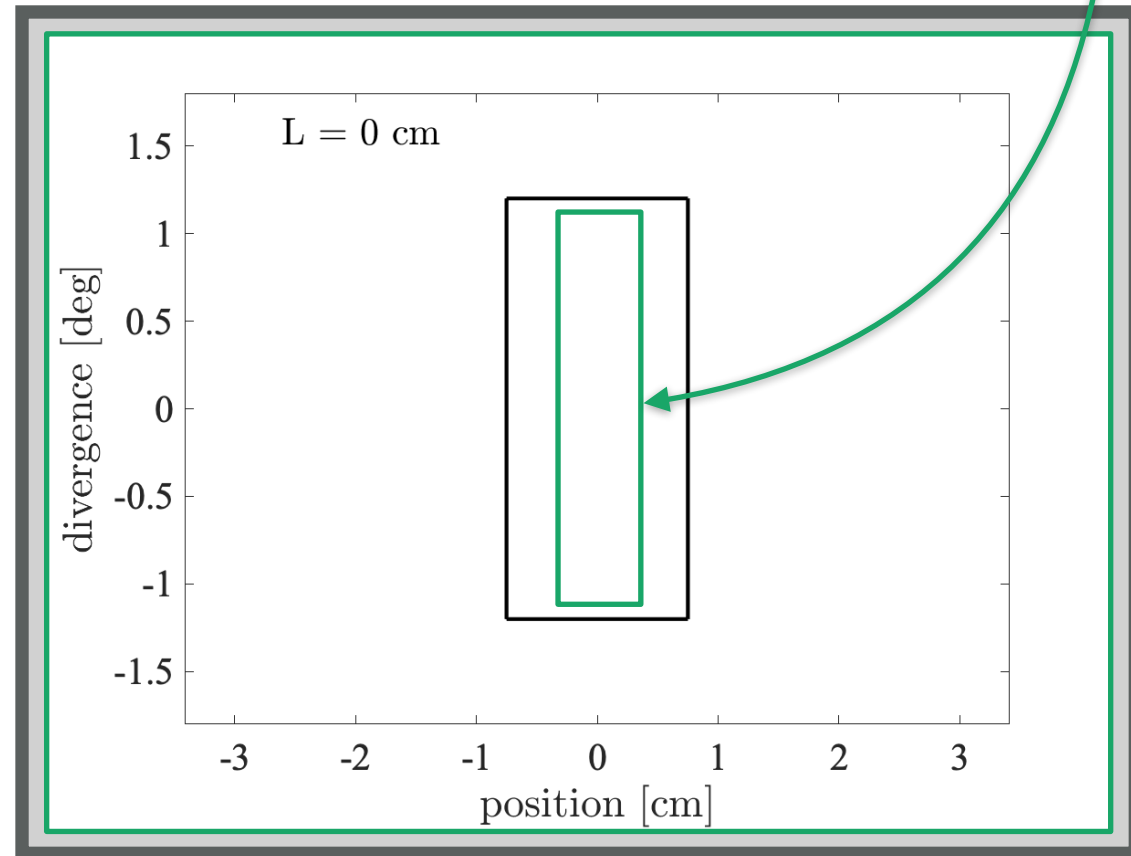
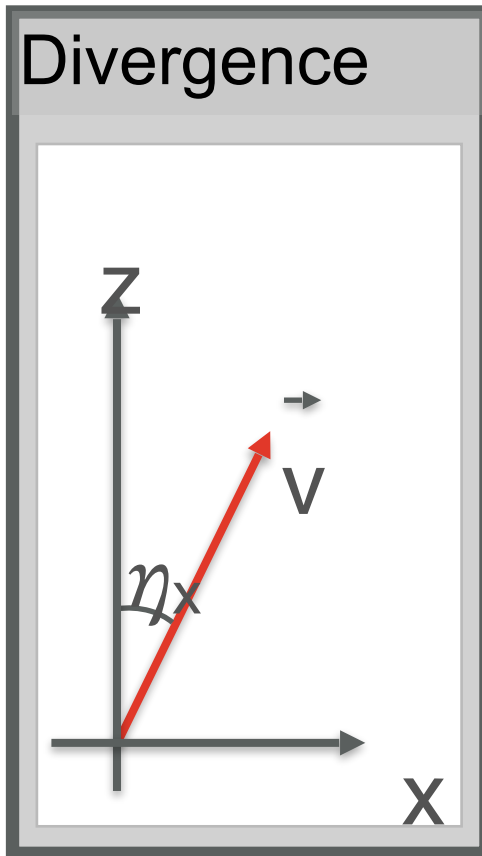
Beam propagation in free space

“Phase-space” at source



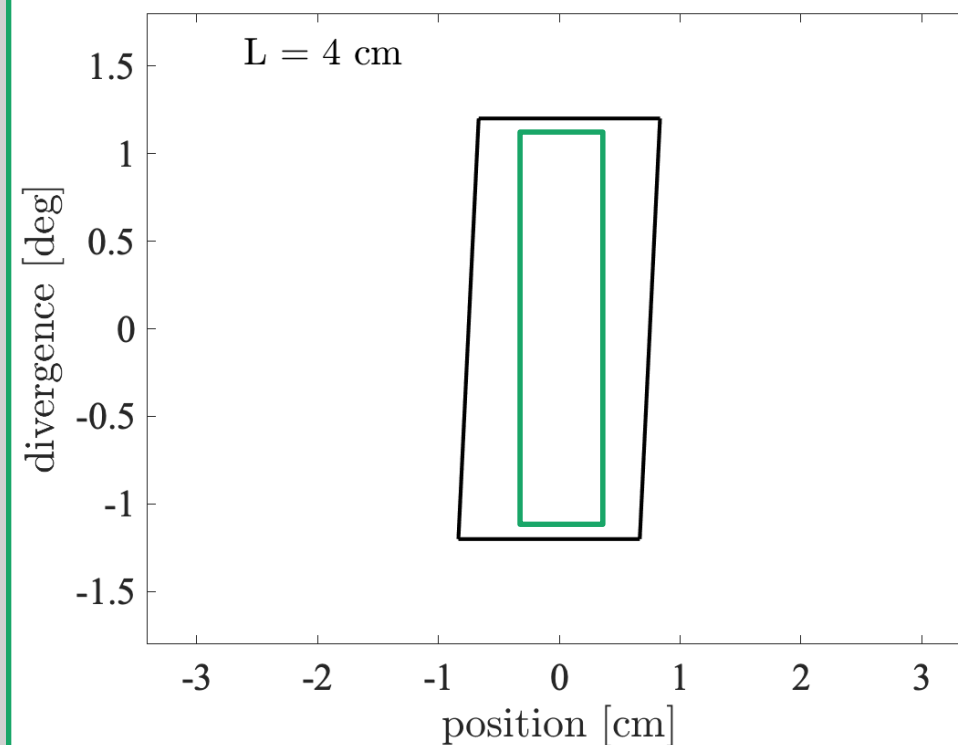
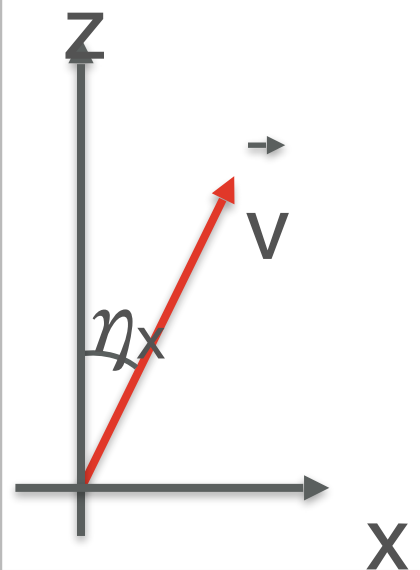
Beam propagation in free space

Wanted “phase-space” at sample



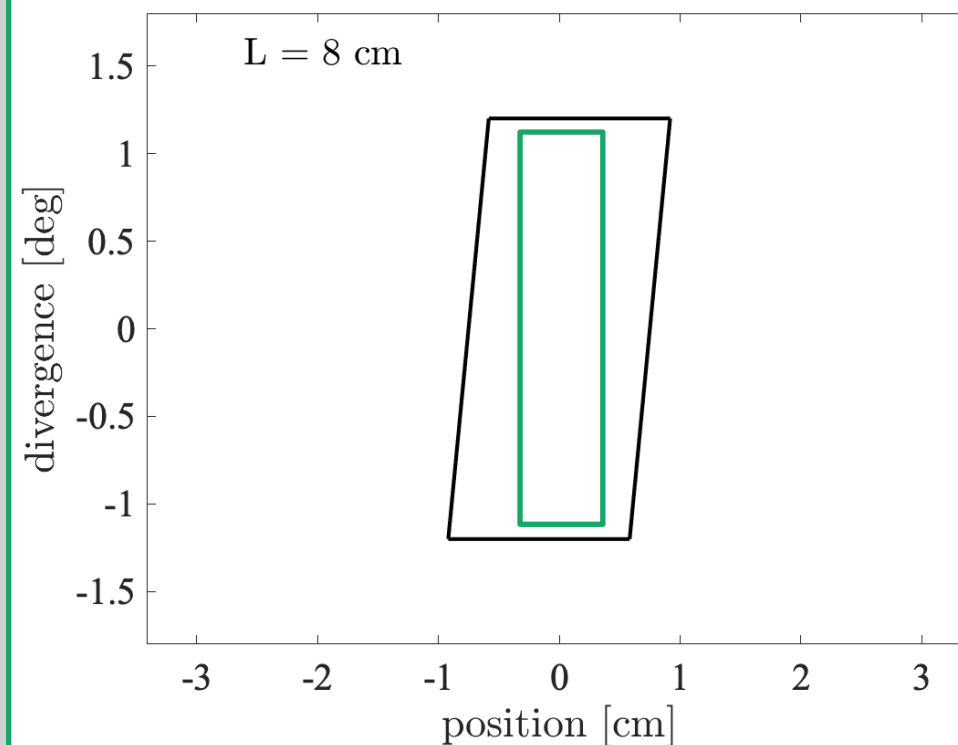
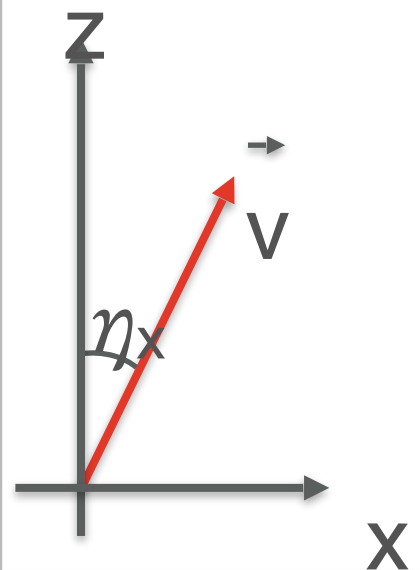
Beam propagation in free space

Divergence



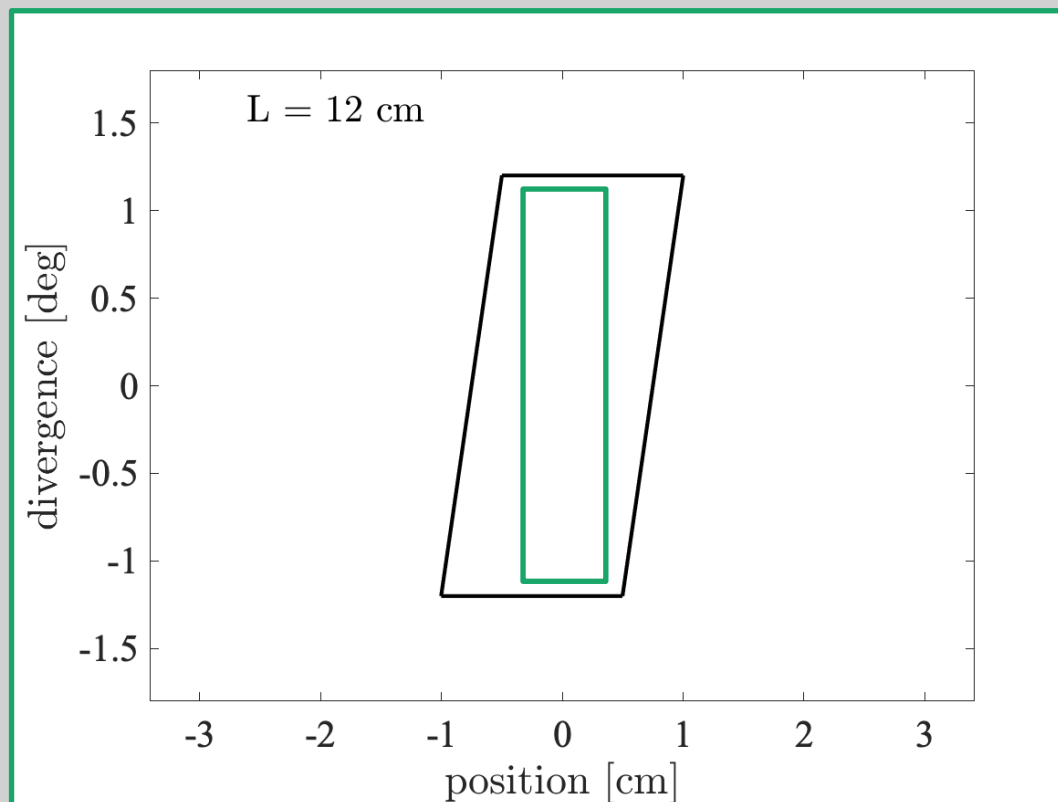
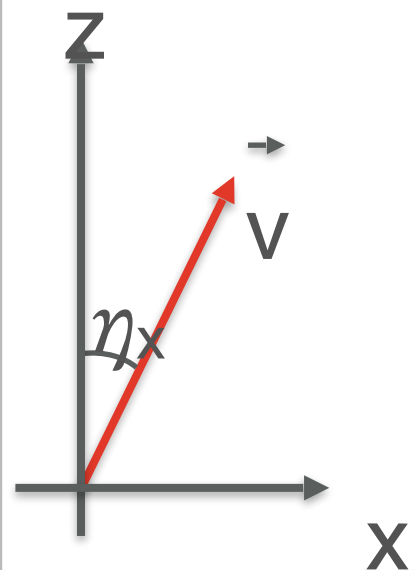
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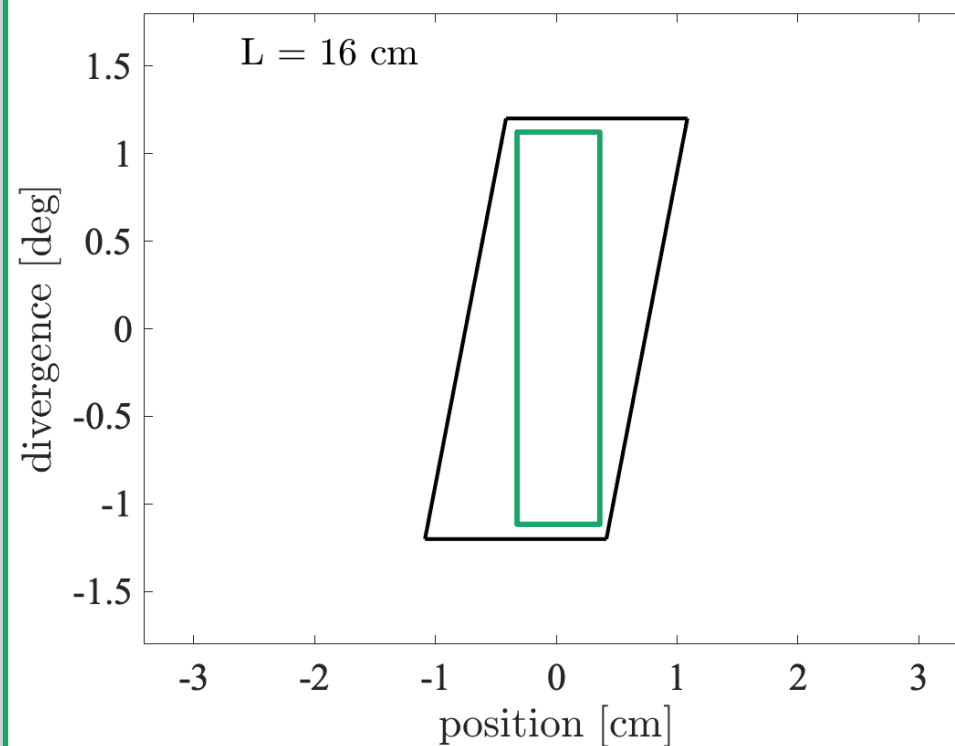
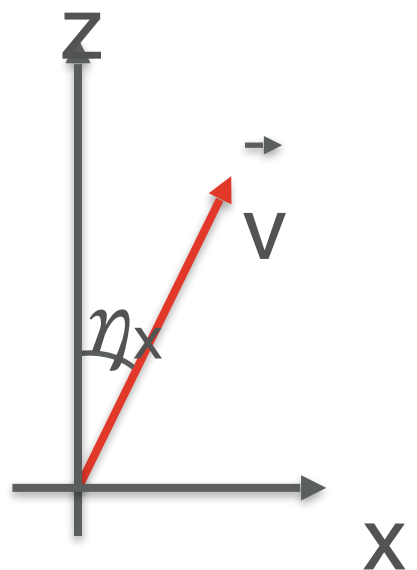
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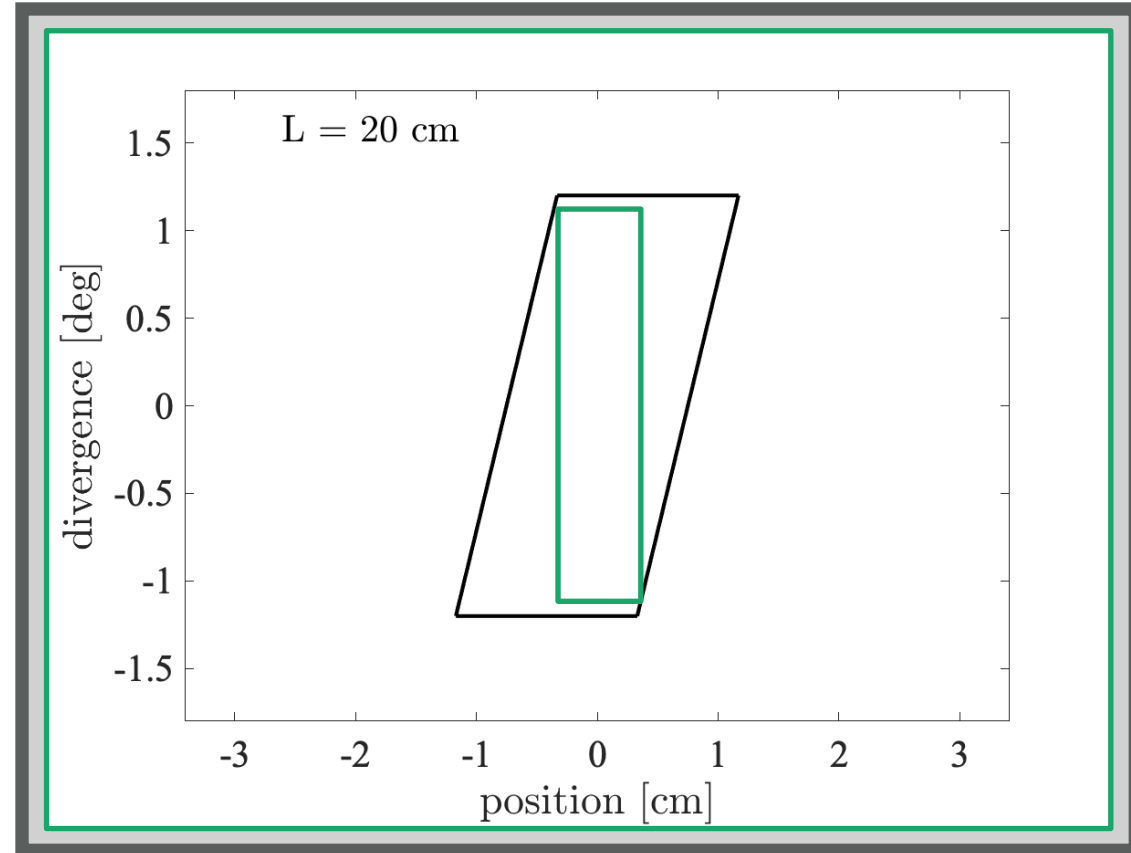
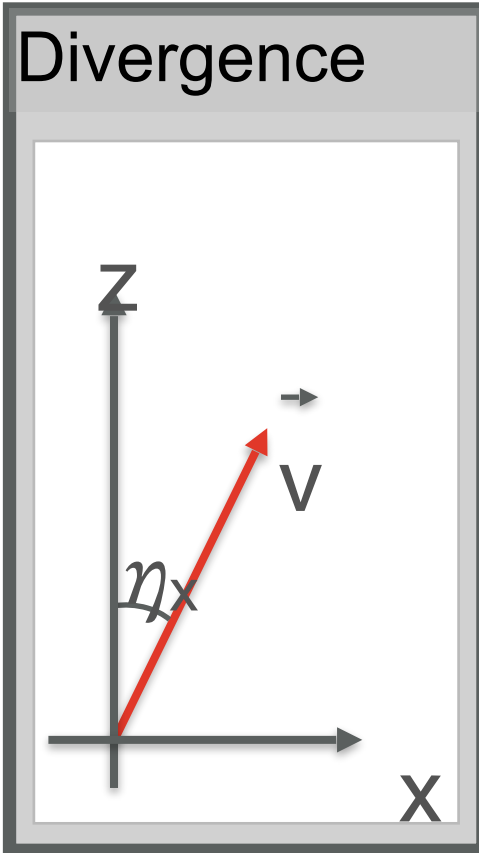


Beam propagation in free space

Divergence

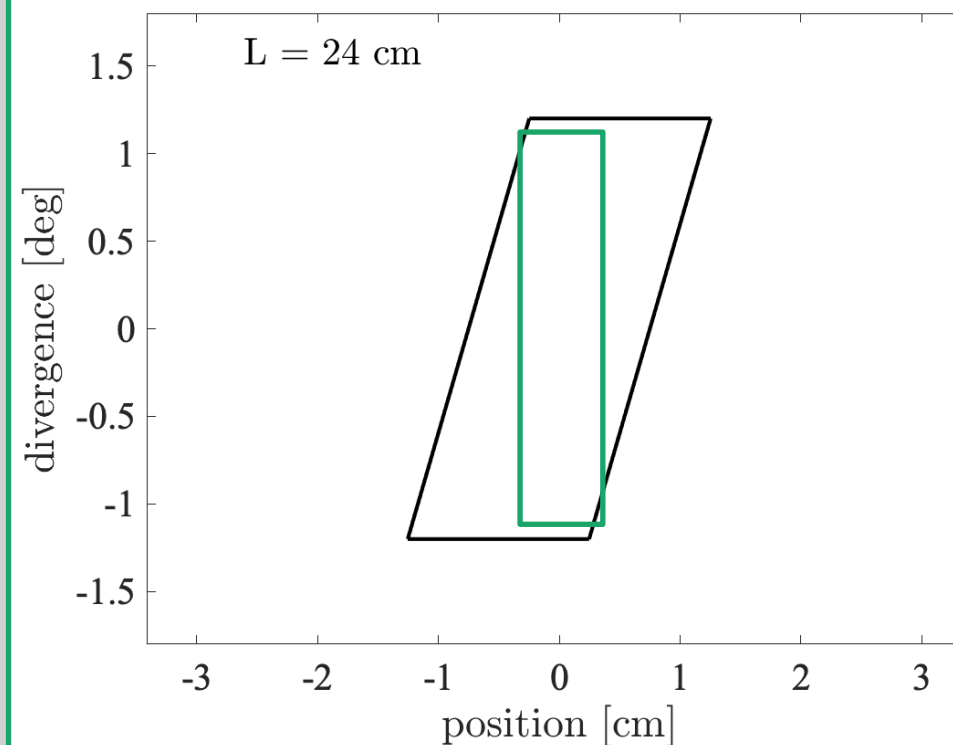
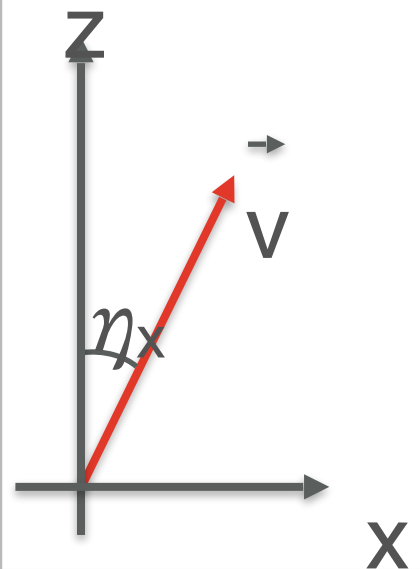


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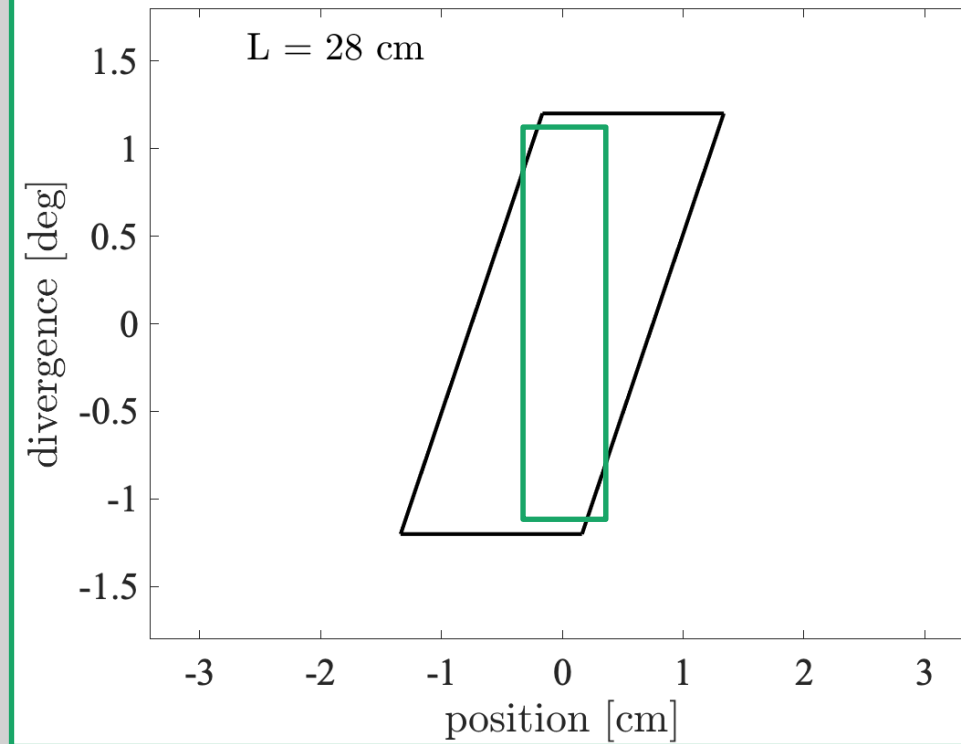
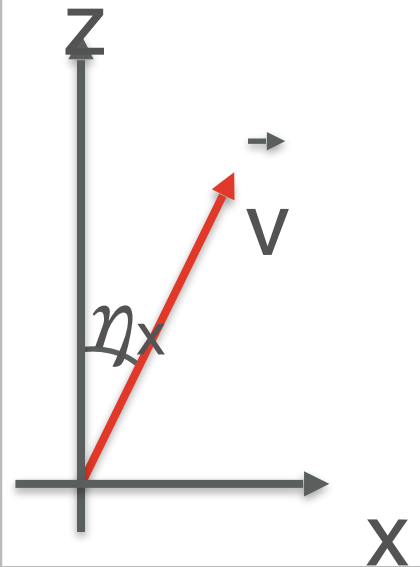
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Divergence



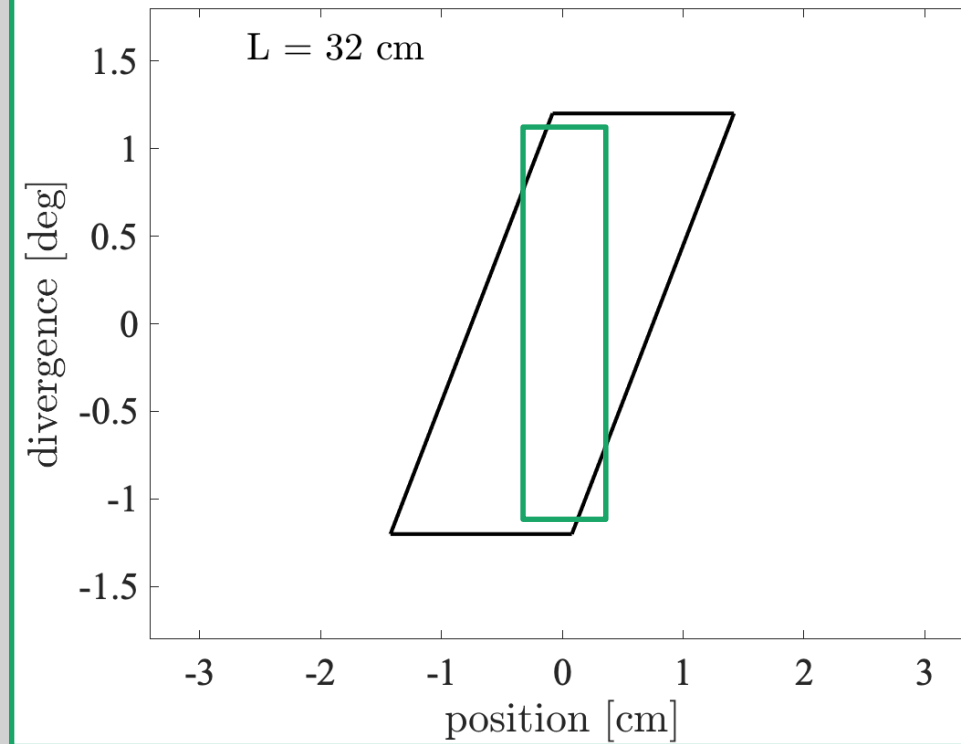
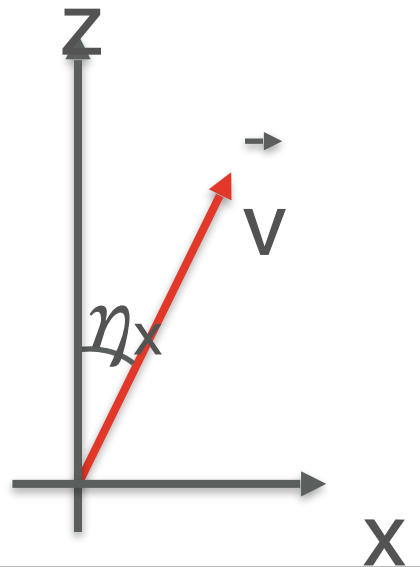
Beam propagation in free space

Divergence



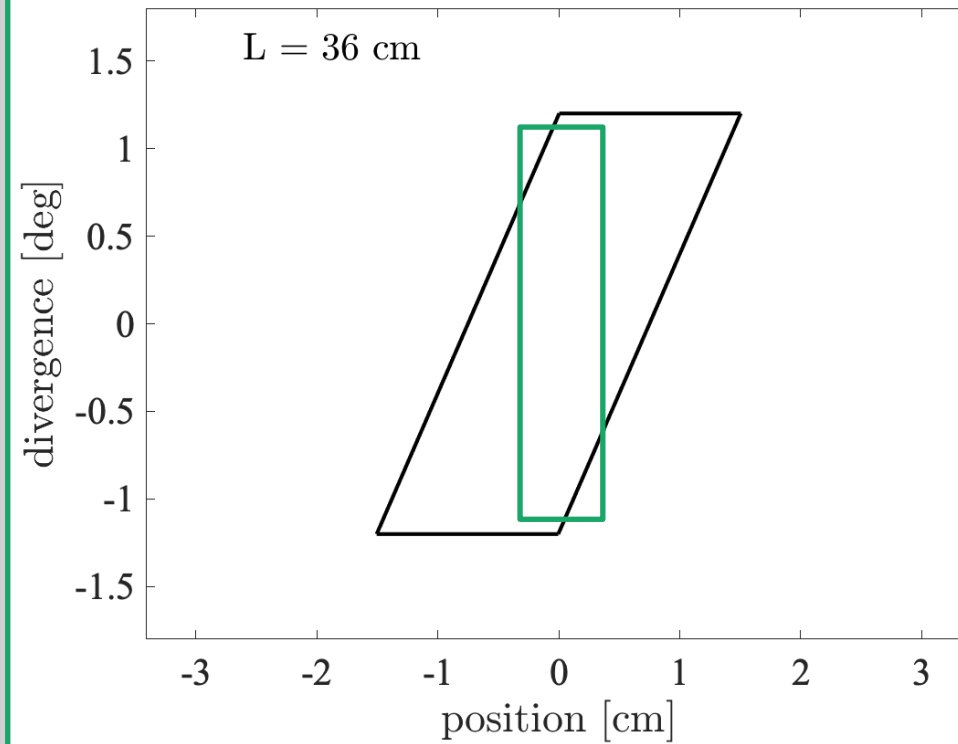
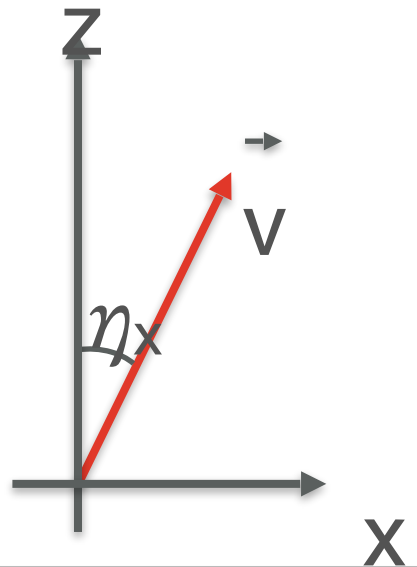
Beam propagation in free space

Divergence



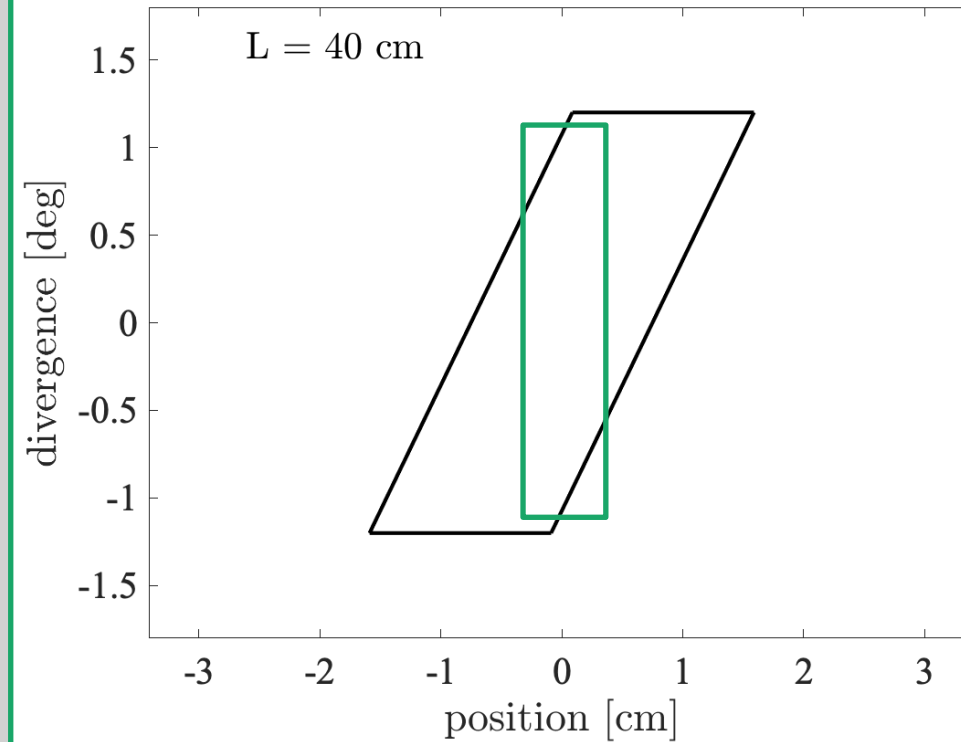
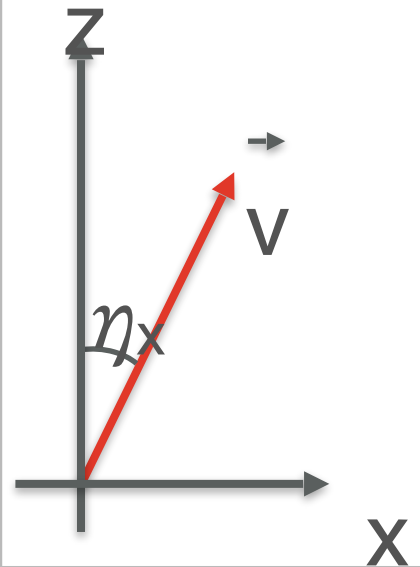
Beam propagation in free space

Divergence



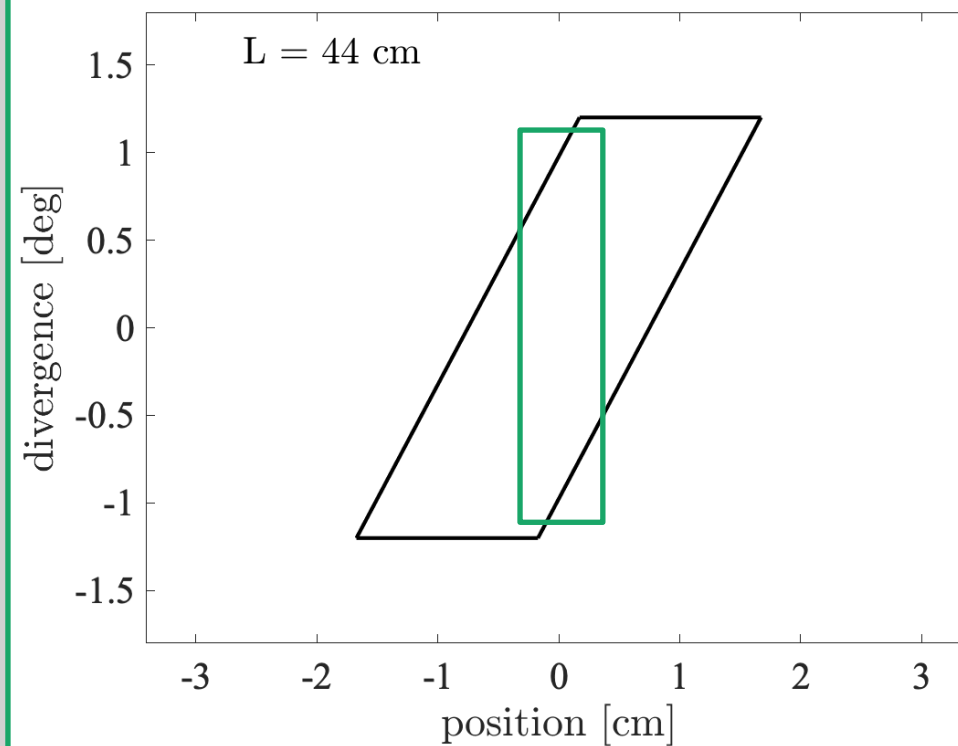
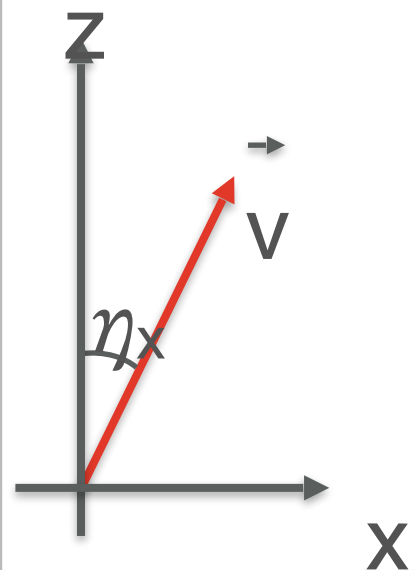
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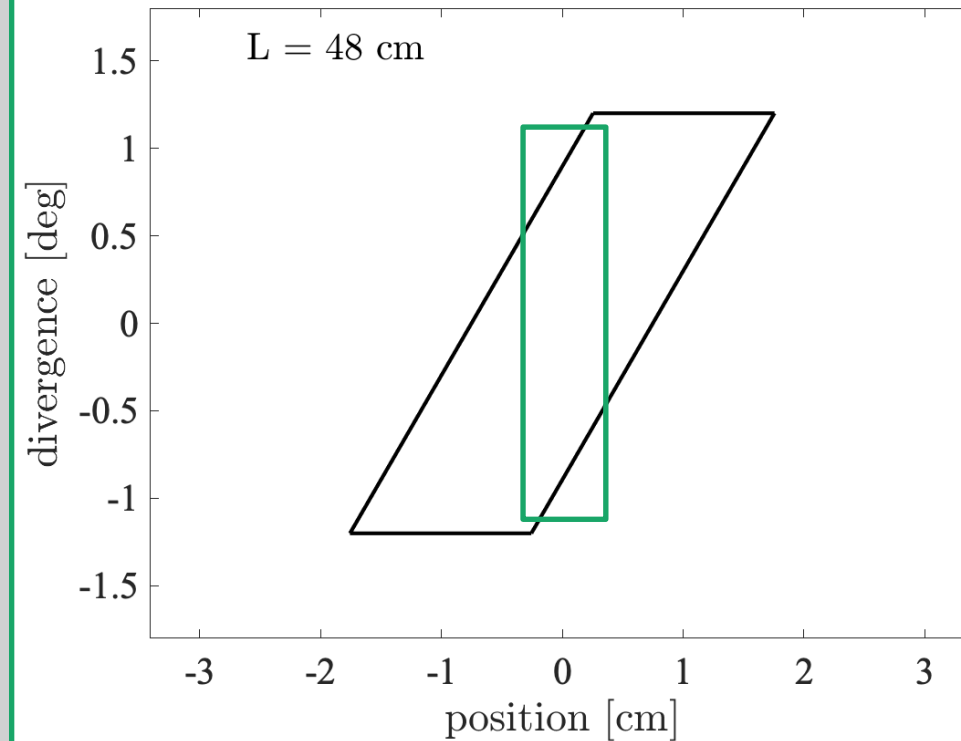
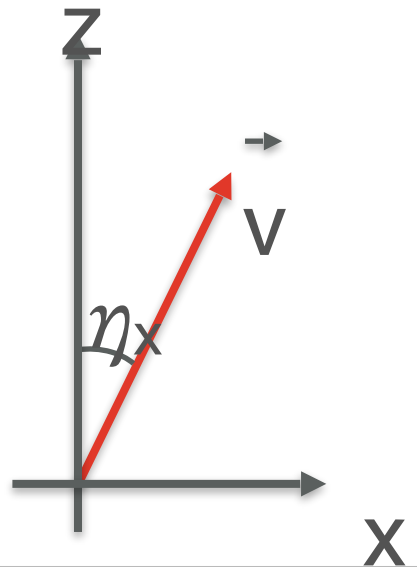
Beam propagation in free space

Divergence



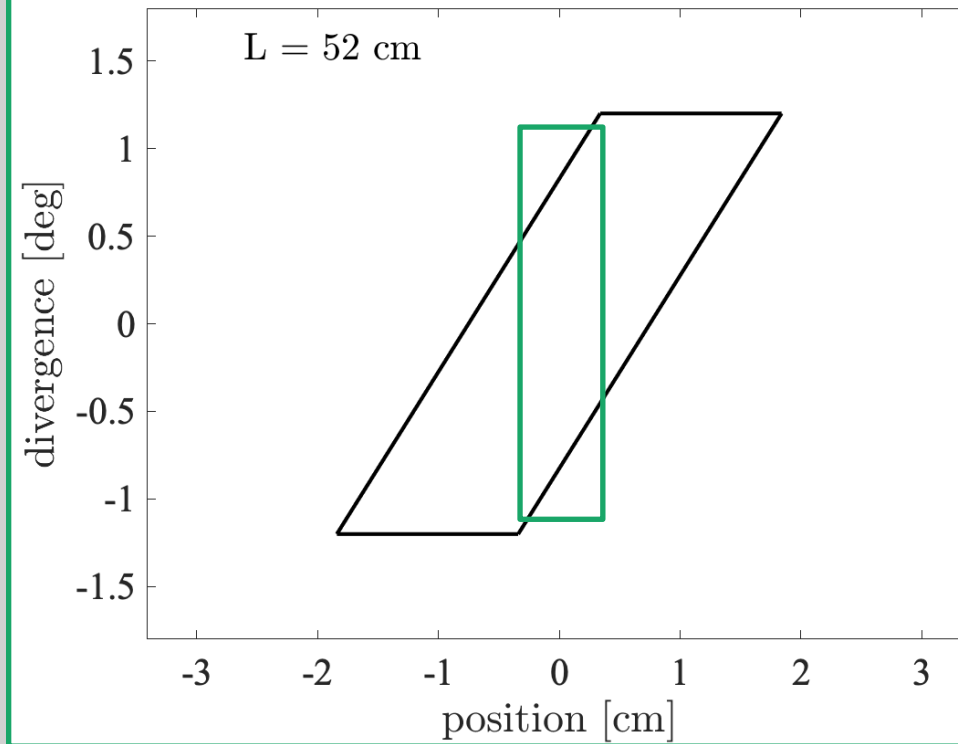
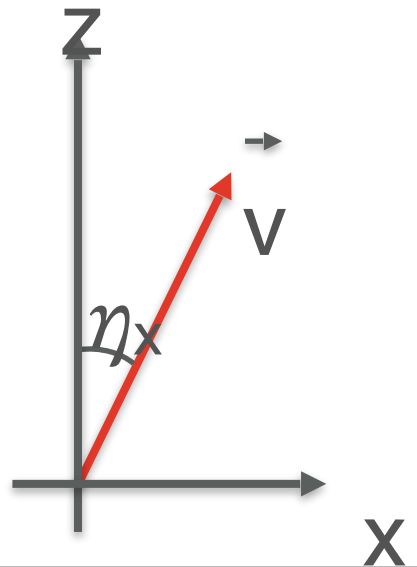
Beam propagation in free space

Divergence



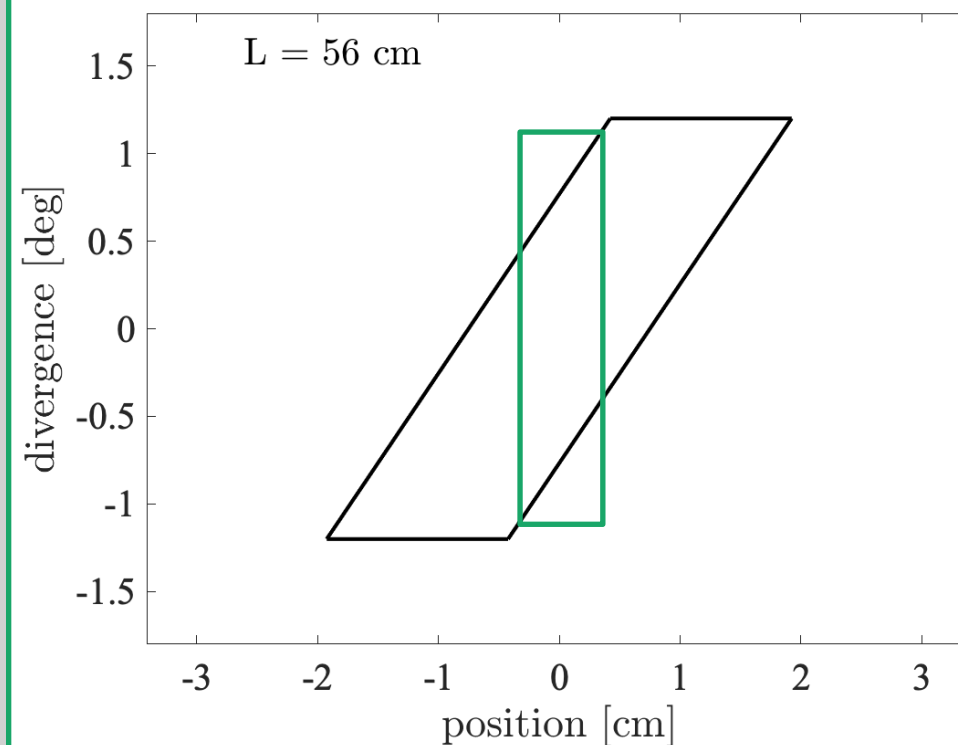
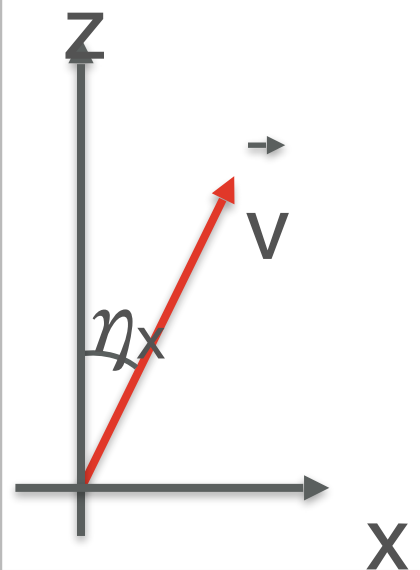
Beam propagation in free space

Divergence



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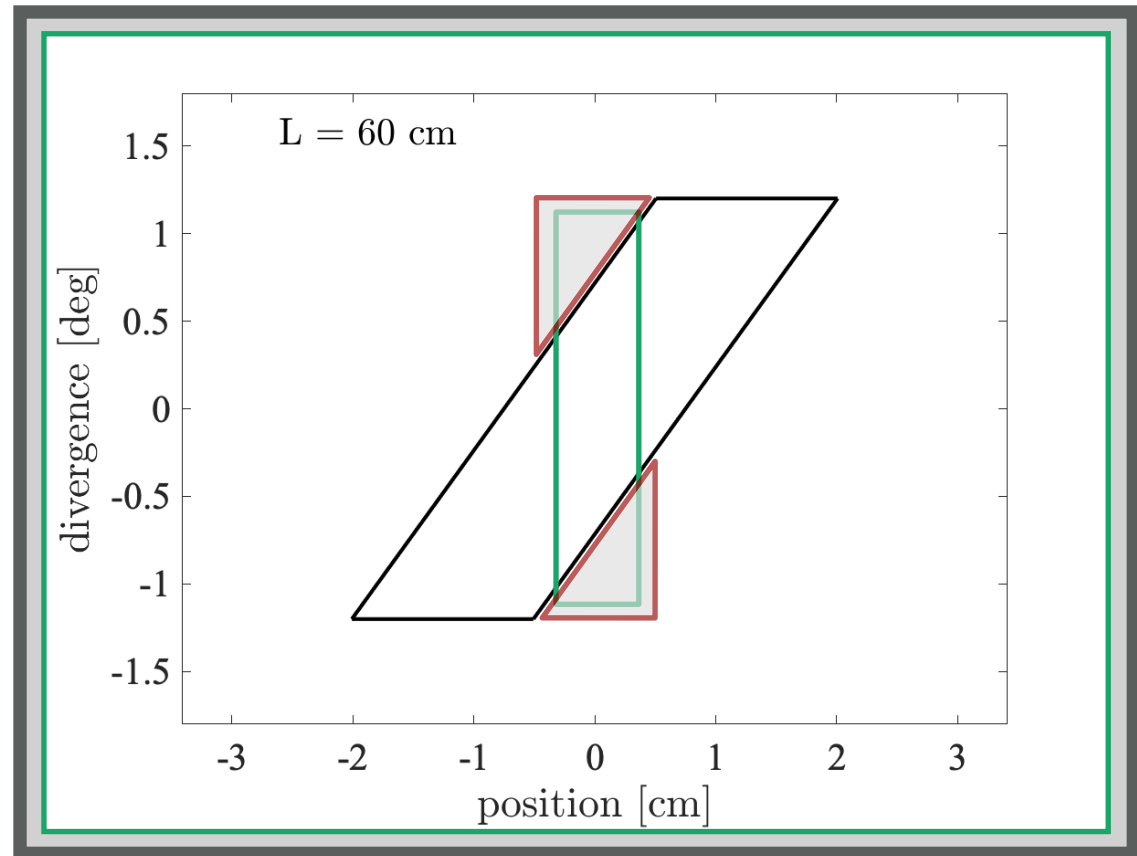
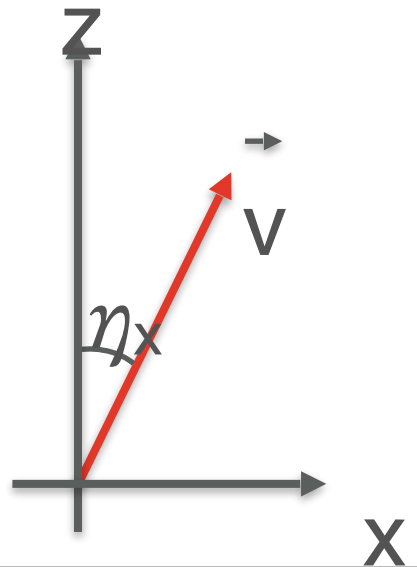
Divergence



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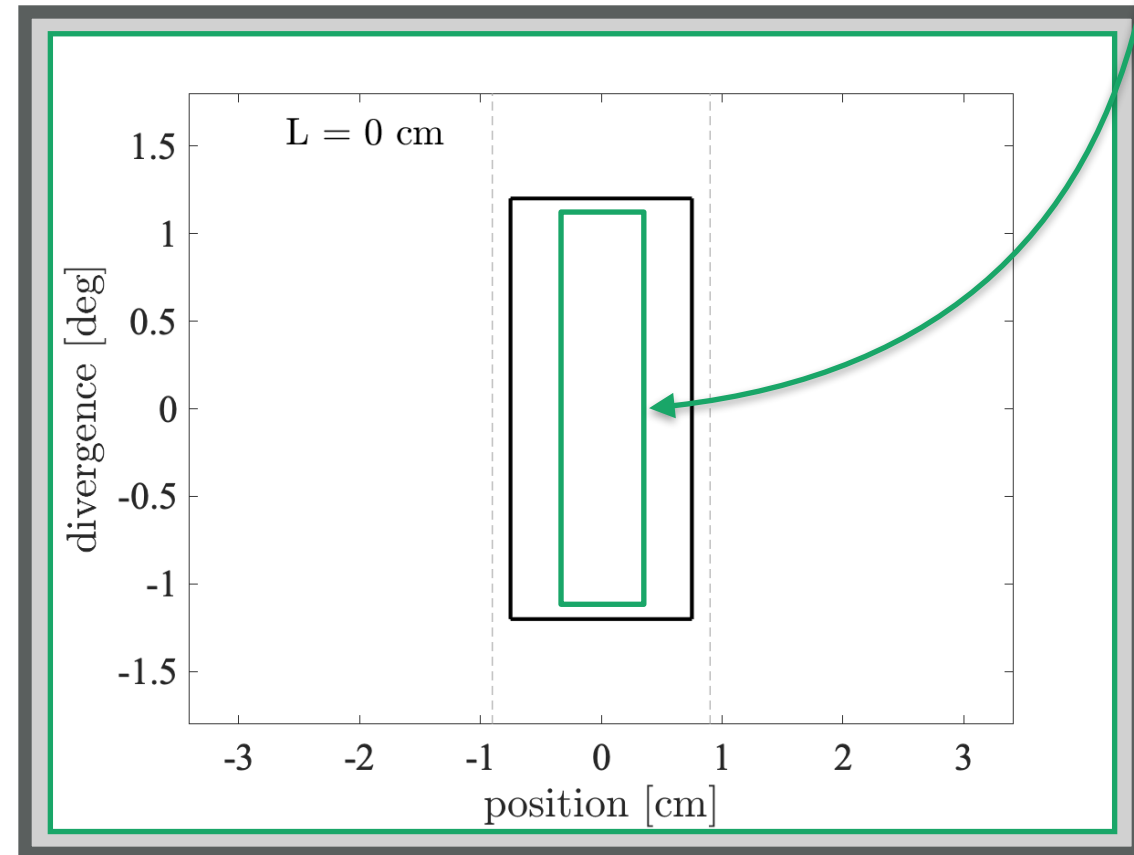
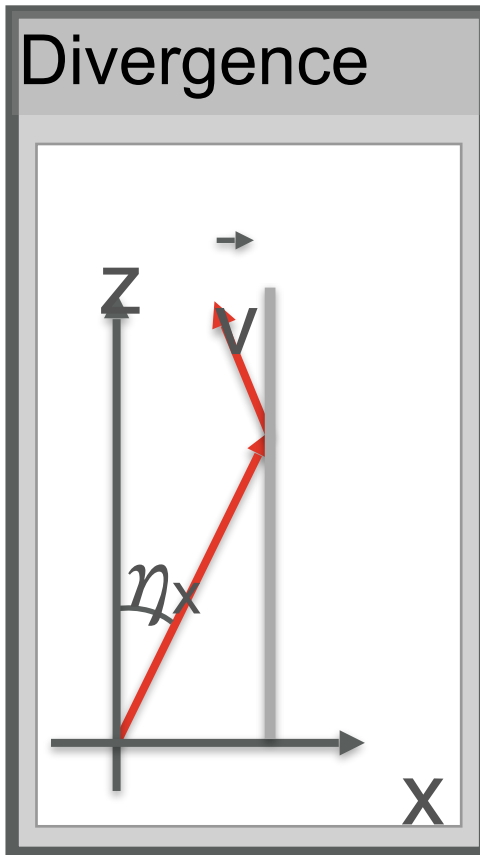
- We lost some phase-space to propagation

Divergence

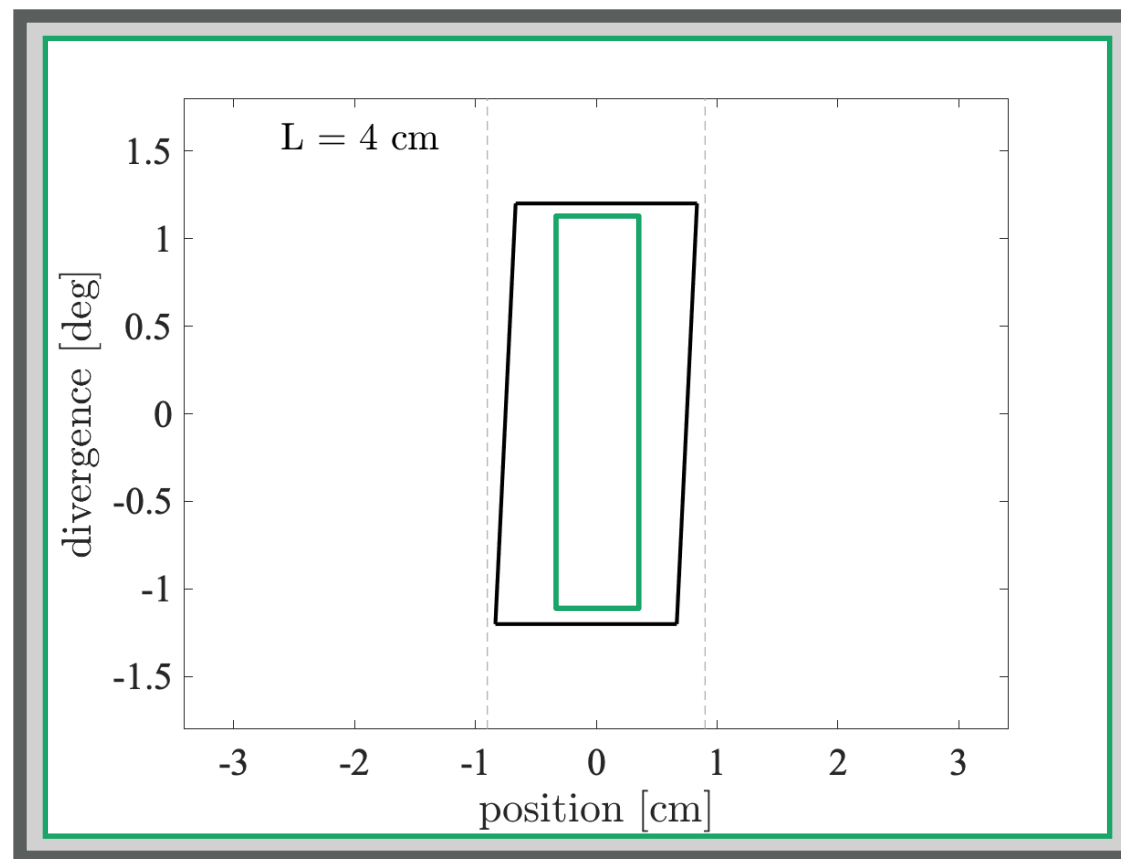
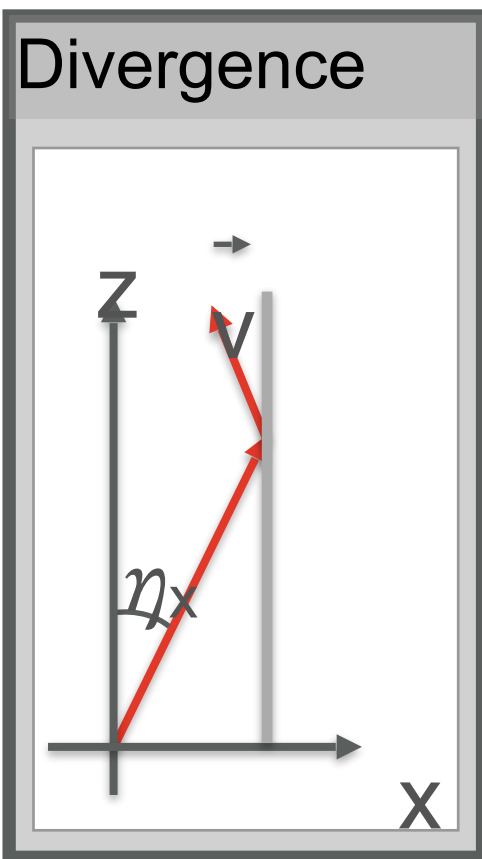


Beam propagation in guide

Wanted “phase-space” at sample

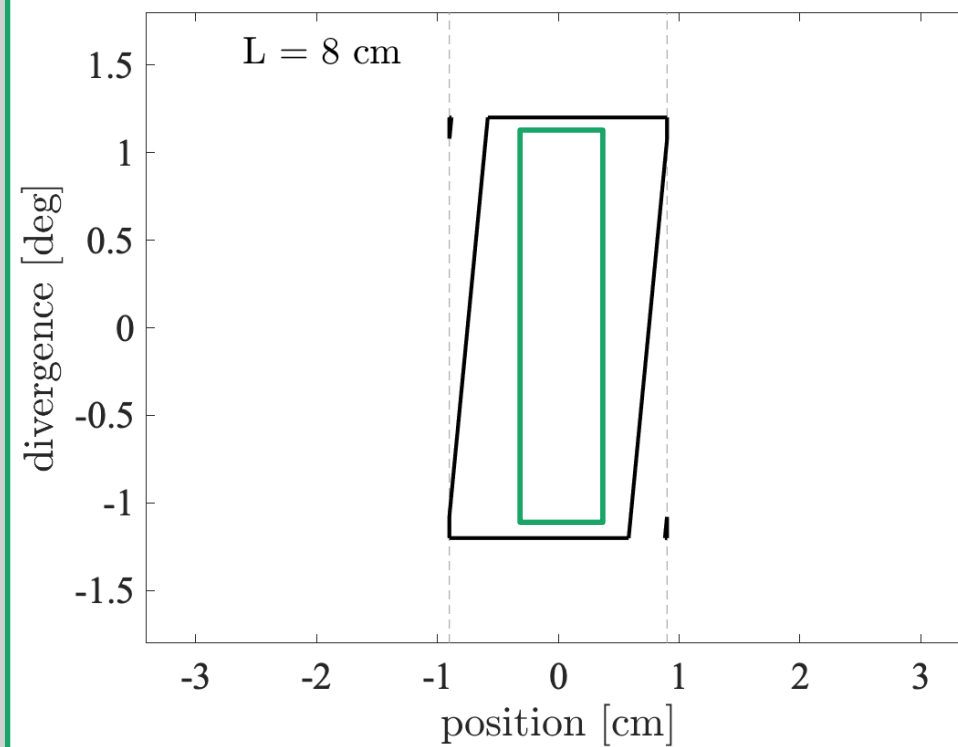
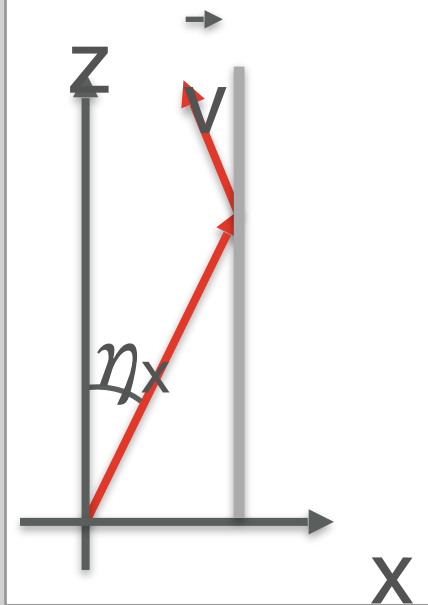


Beam propagation in guide



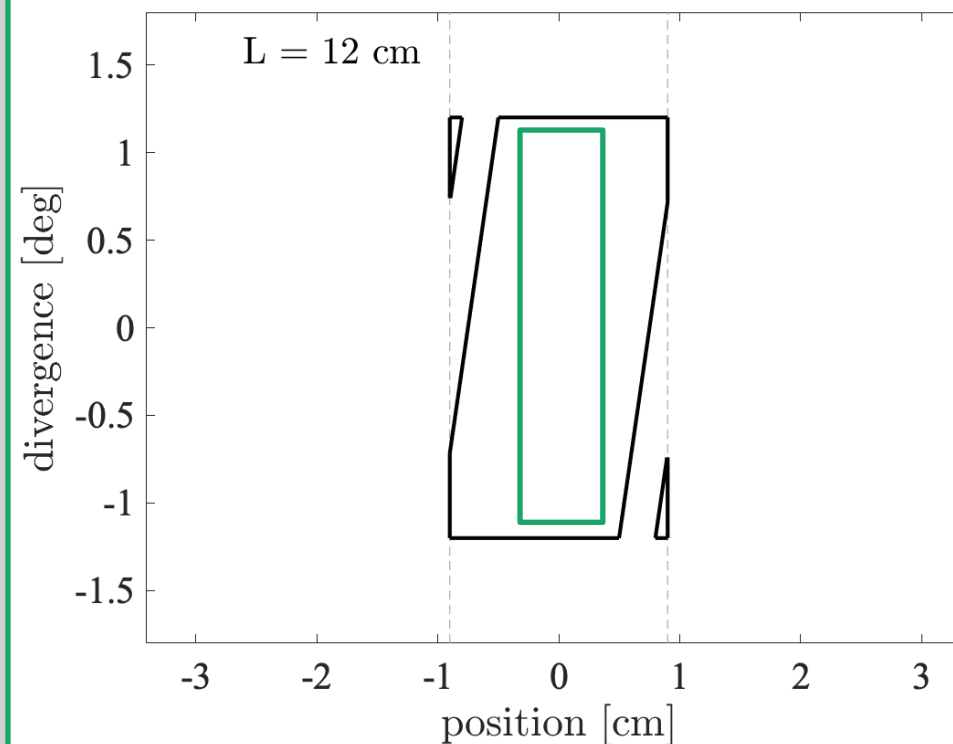
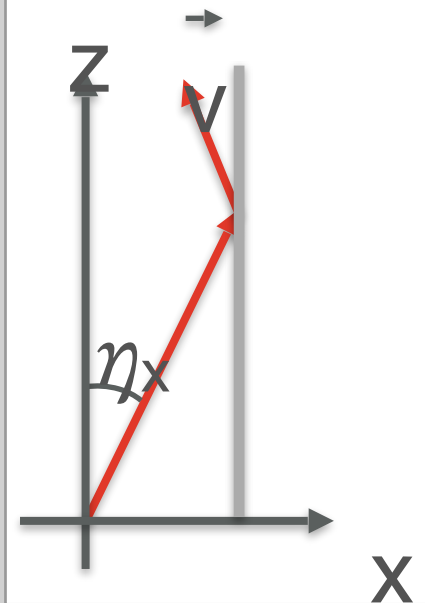
Beam propagation in guide

Divergence



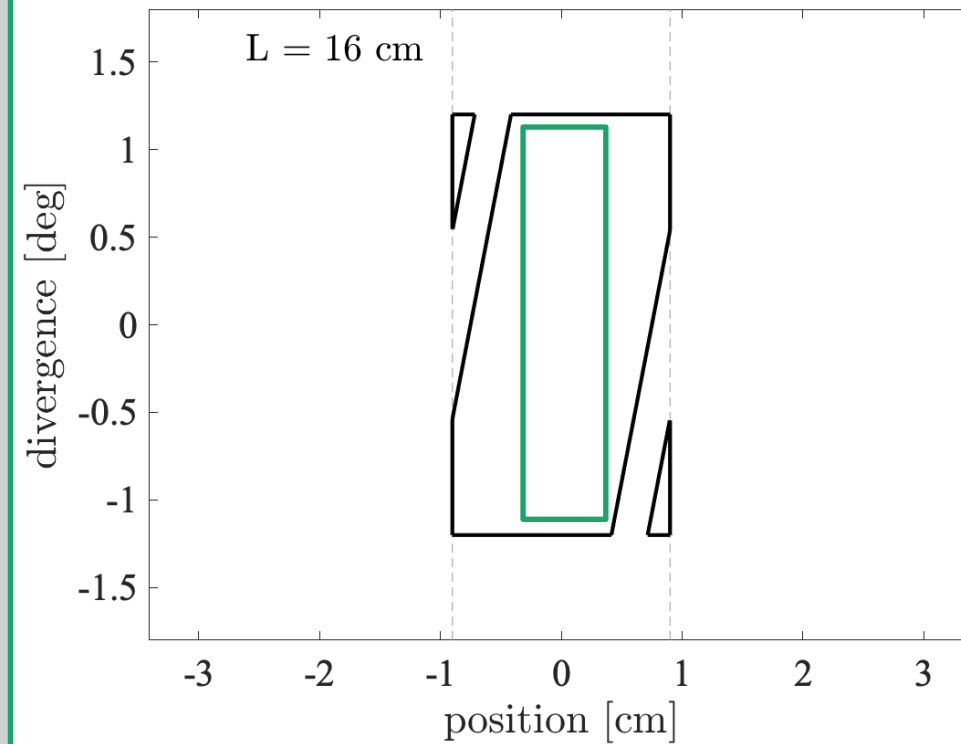
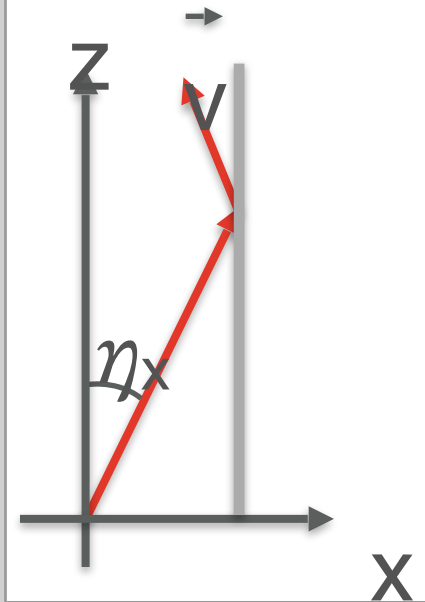
Beam propagation in guide

Divergence

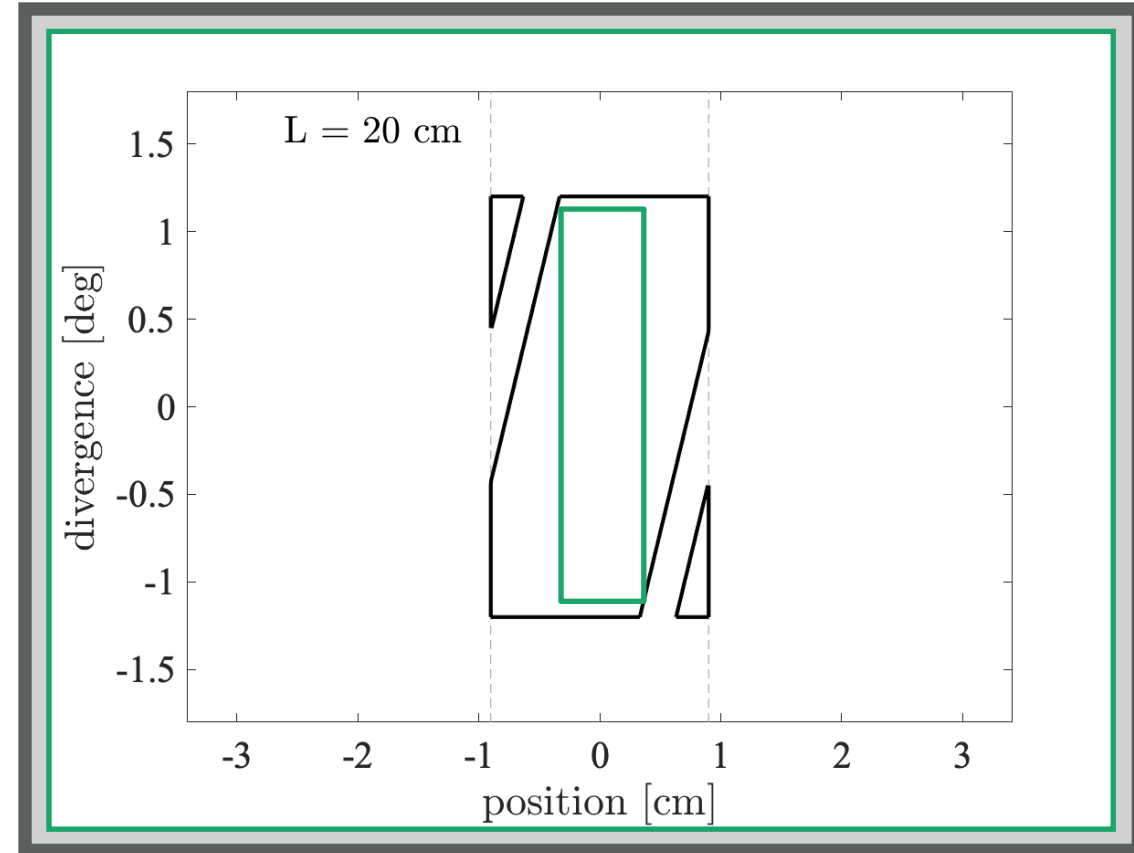
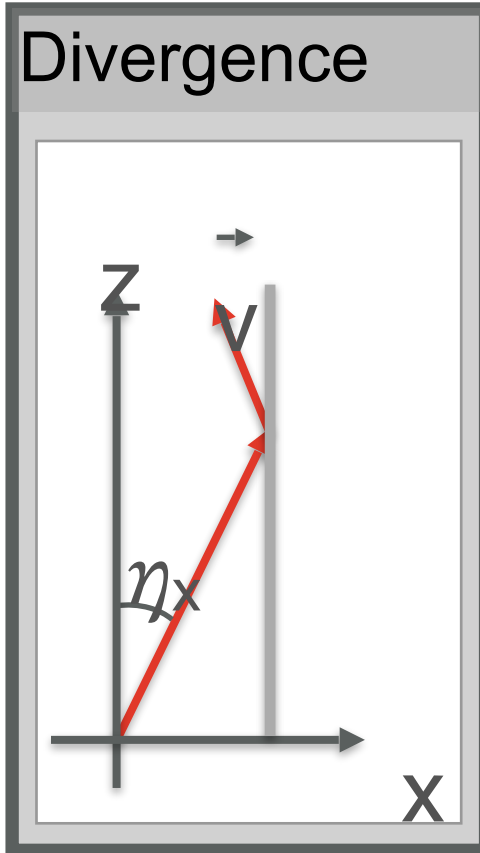


Beam propagation in guide

Divergence

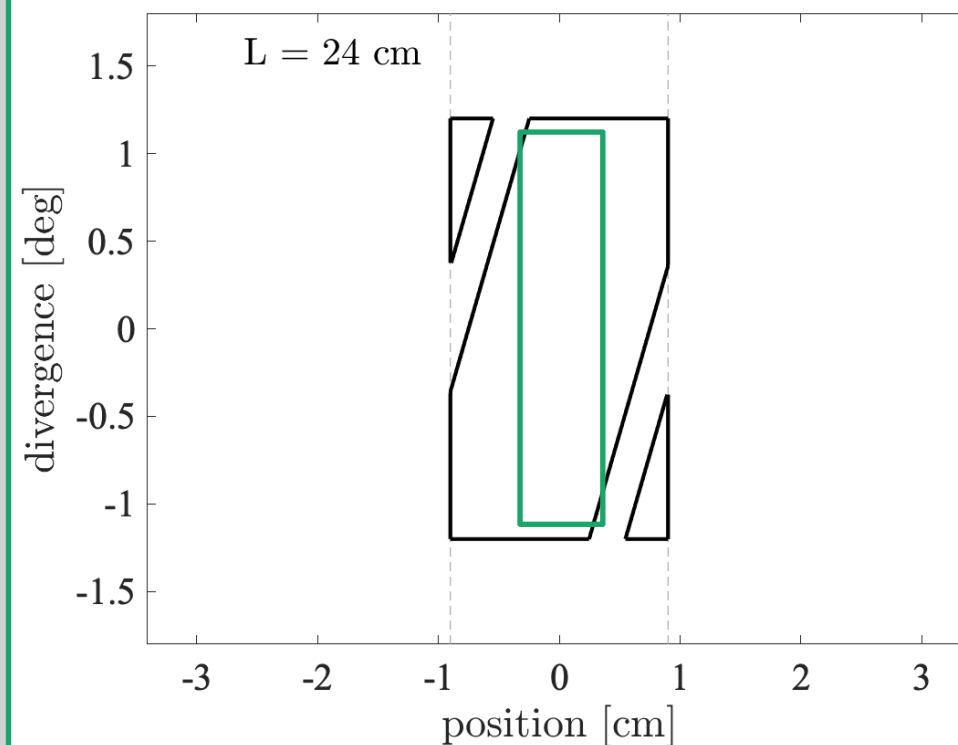
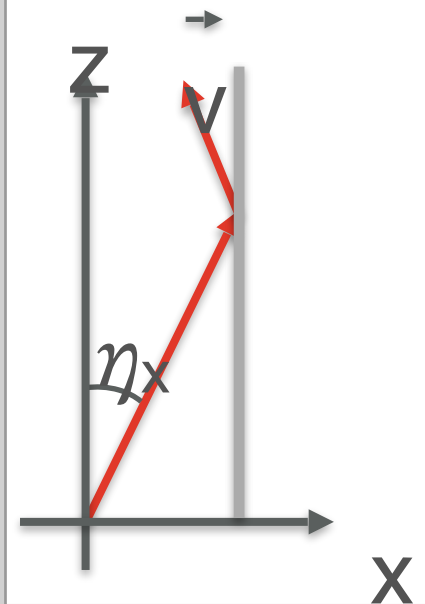


Beam propagation in guide



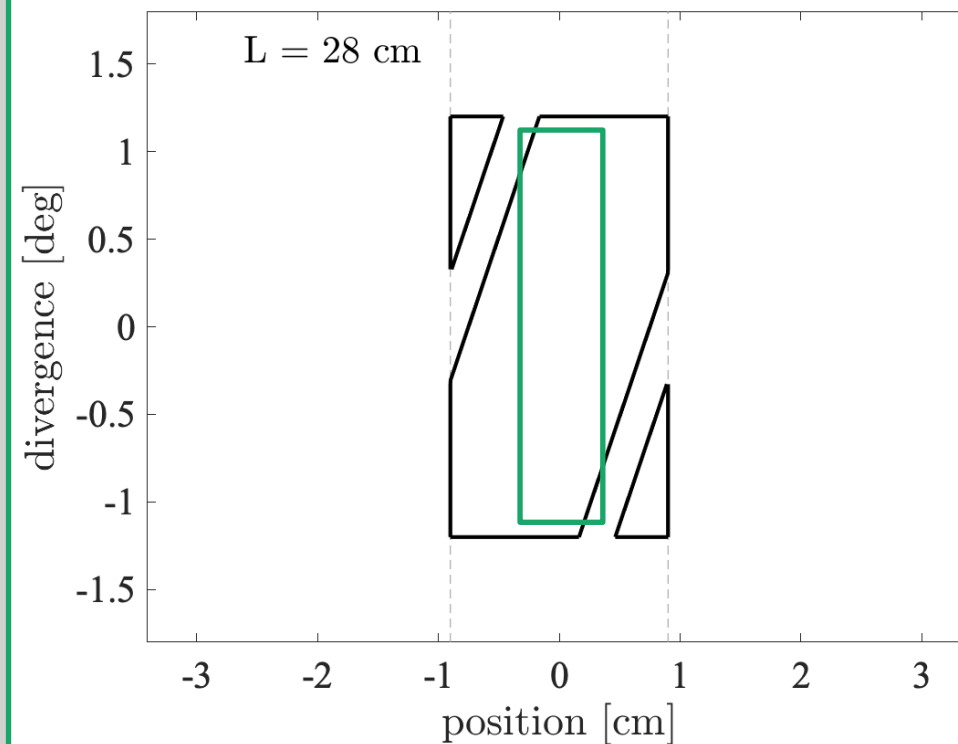
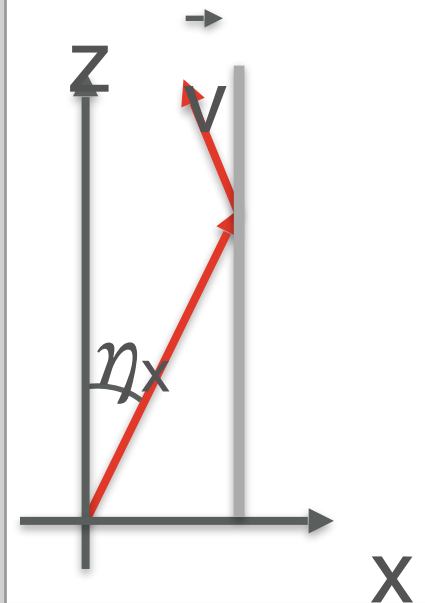
Beam propagation in guide

Divergence



Beam propagation in guide

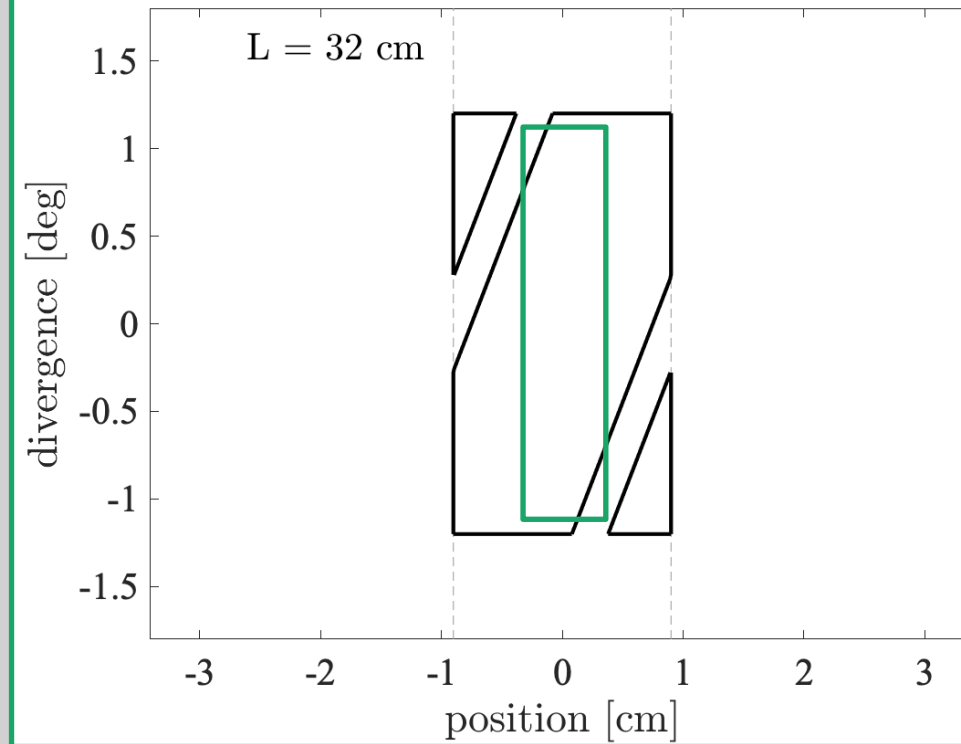
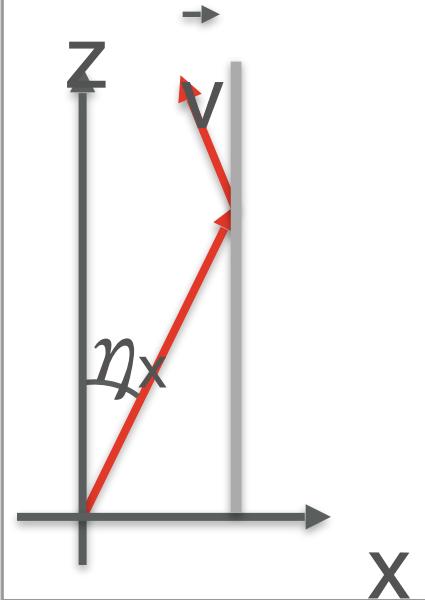
Divergence





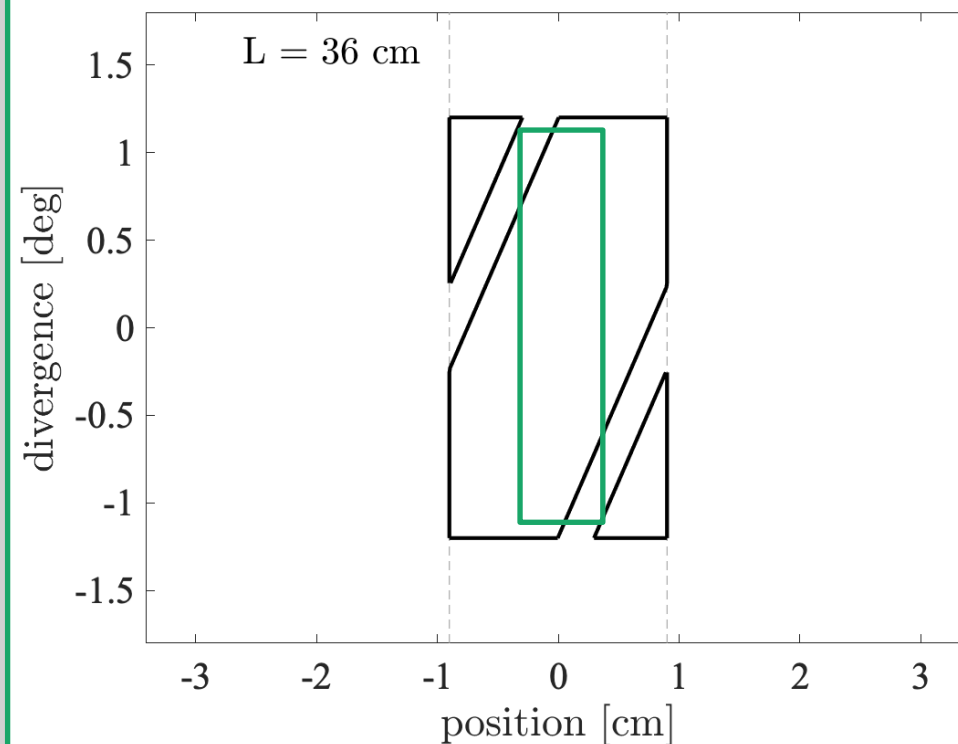
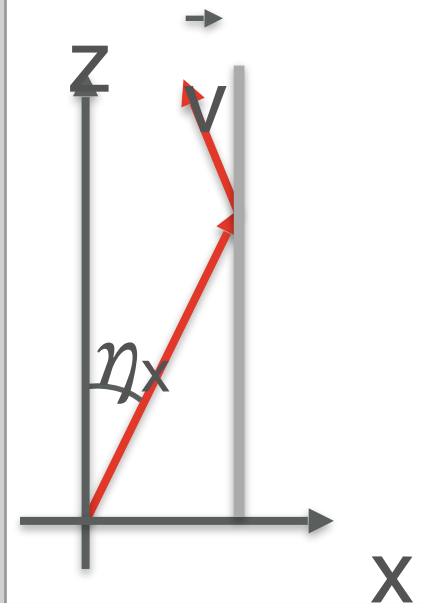
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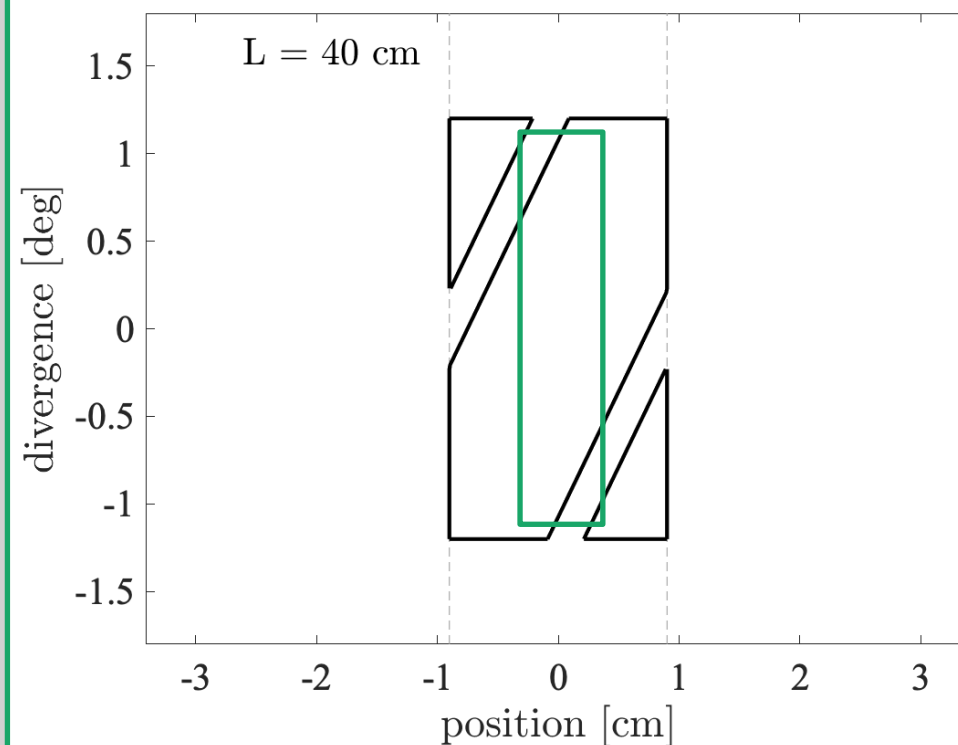
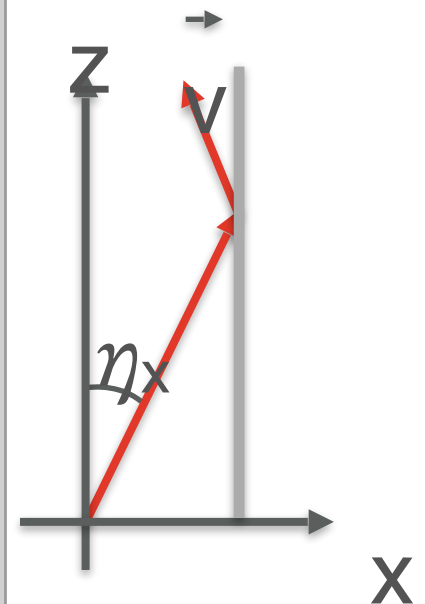
Beam propagation in guide

Divergence



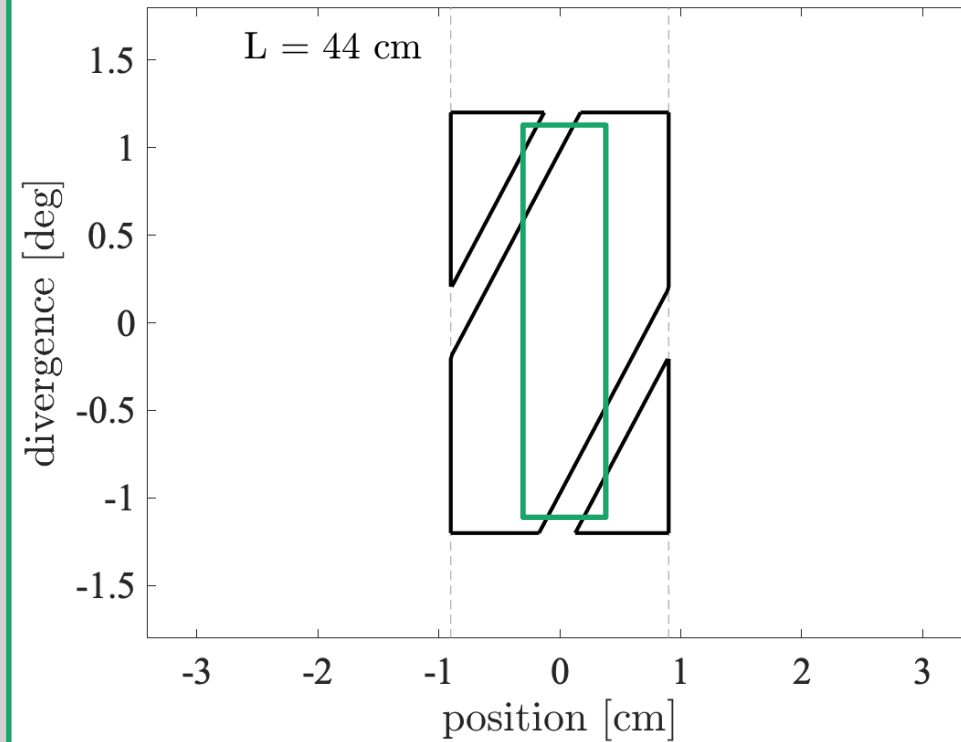
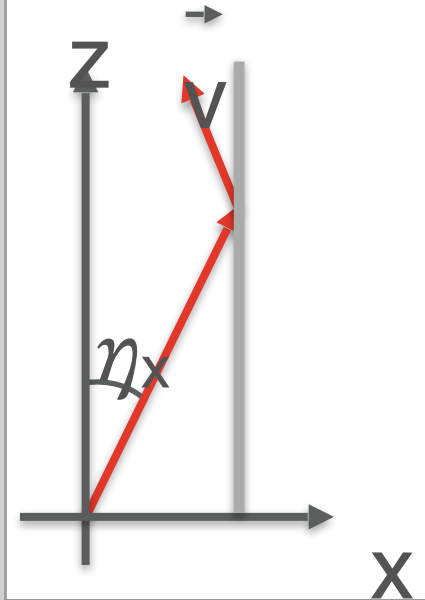
Beam propagation in guide

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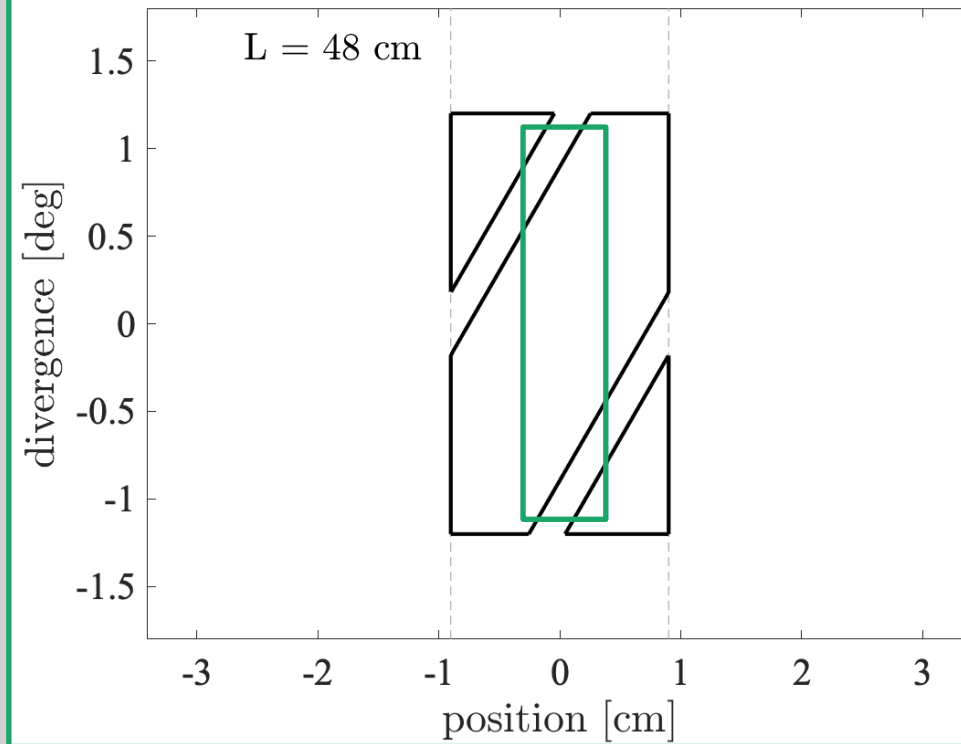
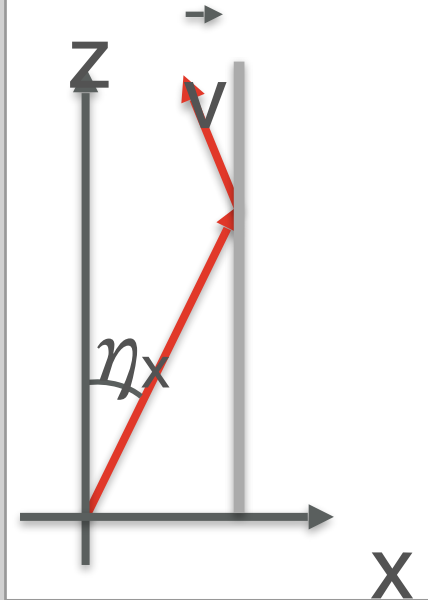
Divergence





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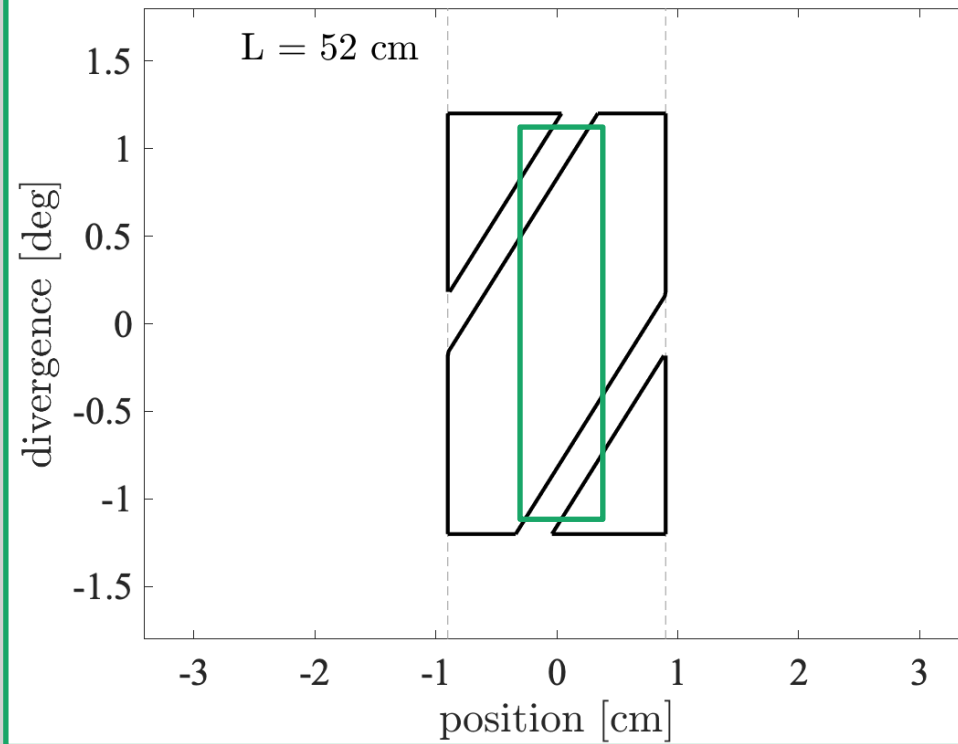
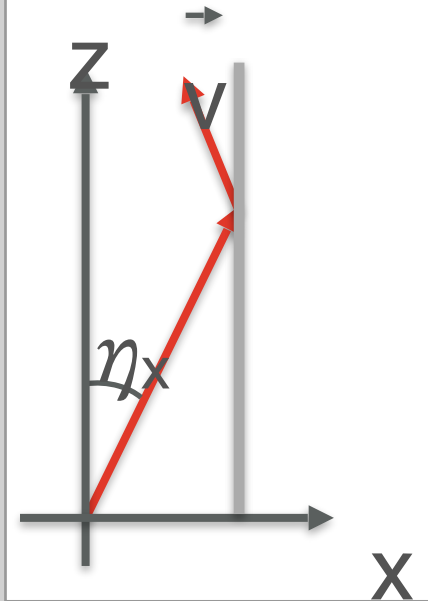
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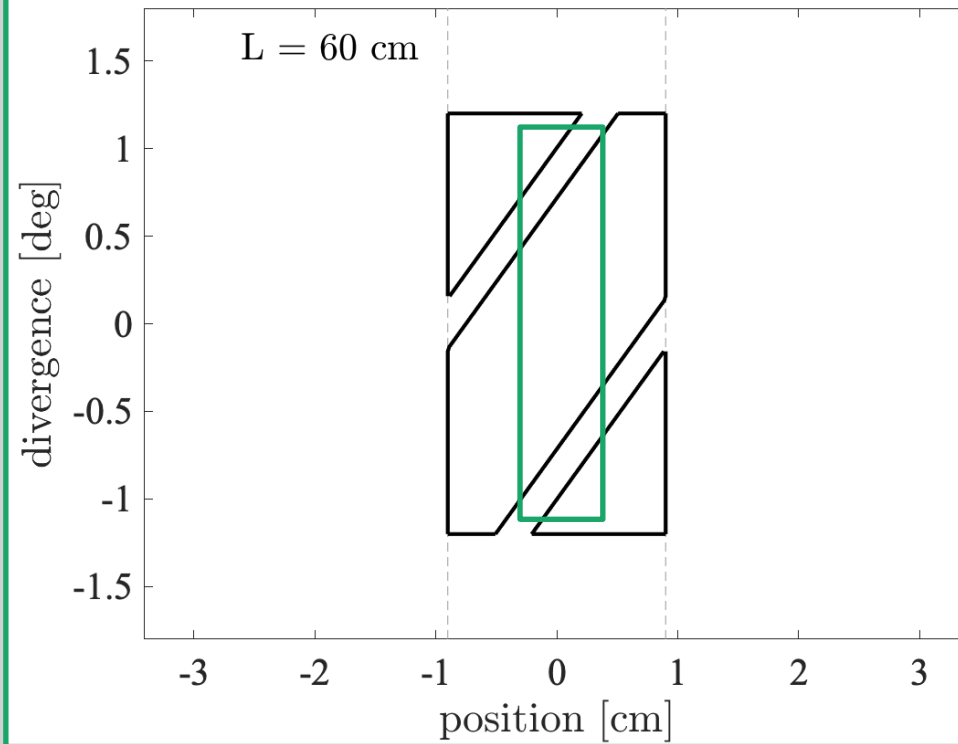
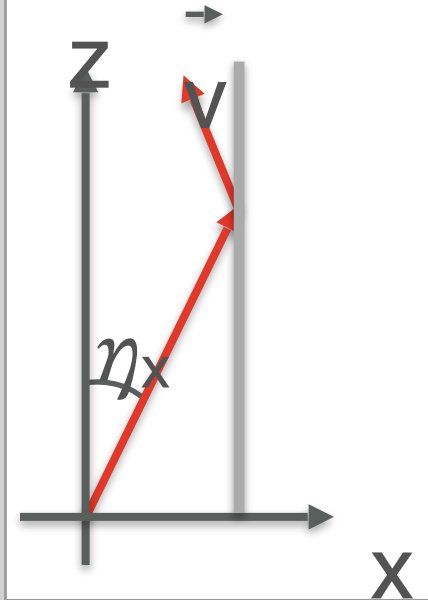
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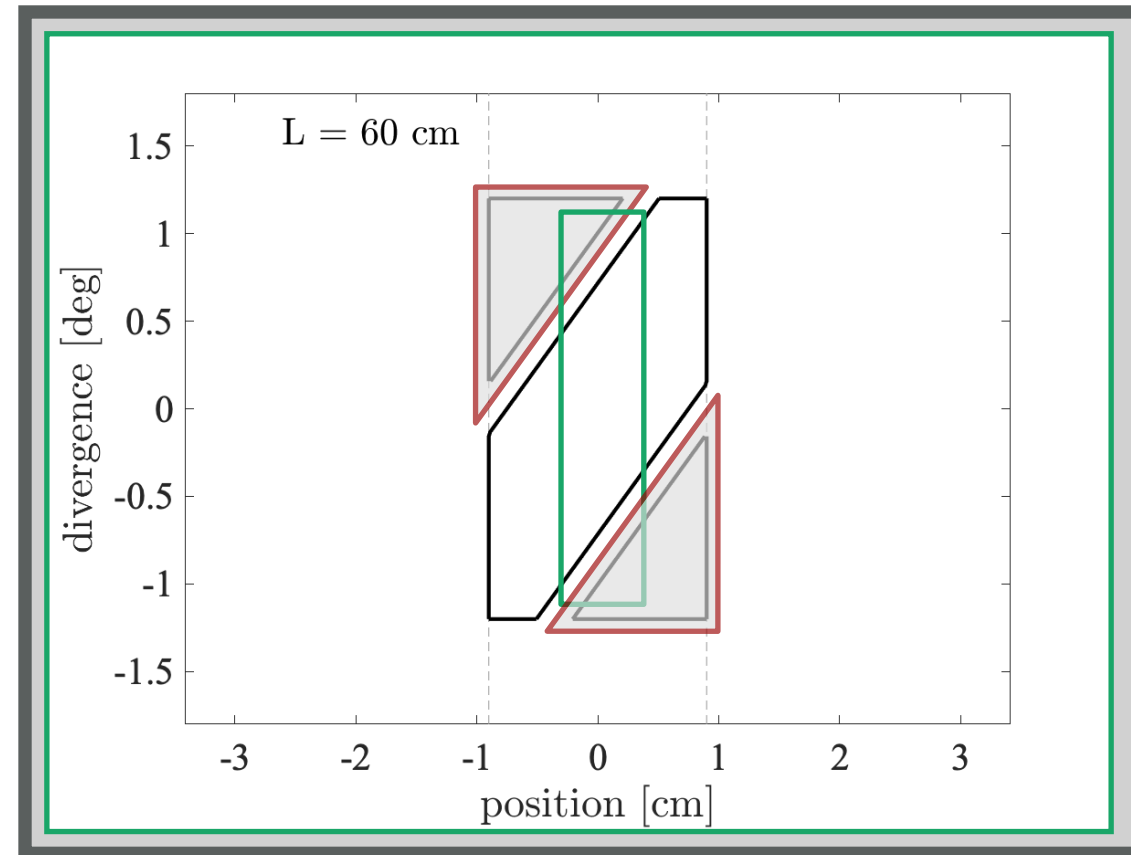
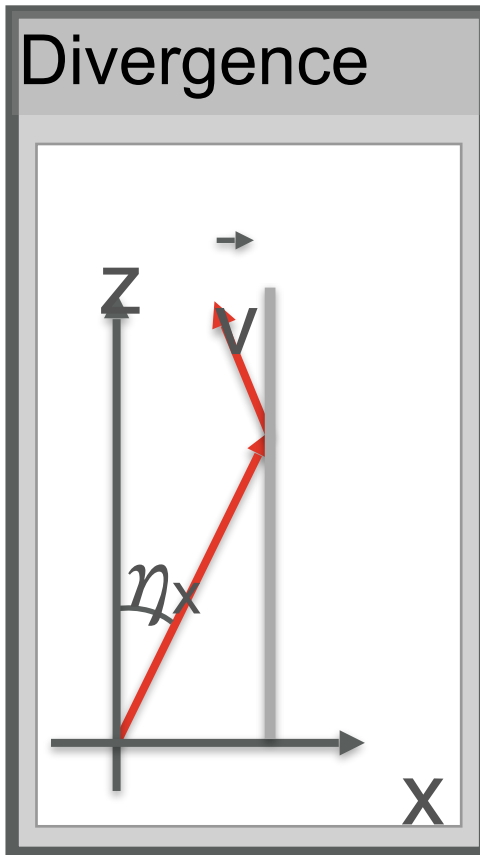
Beam propagation in guide

Divergence



Beam propagation in guide

We got some phase-space back from guide reflection!

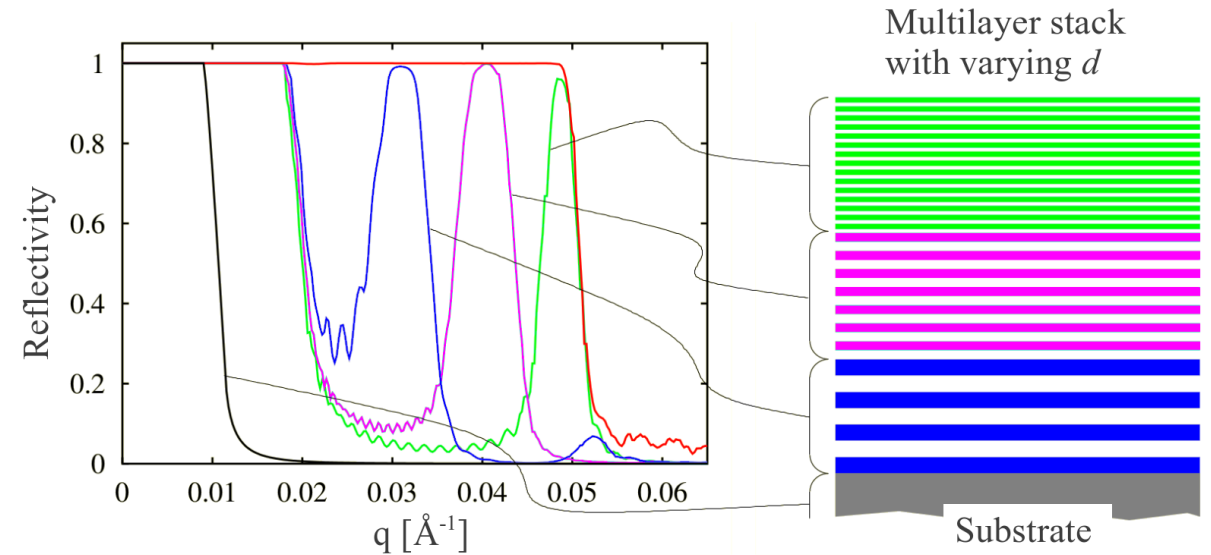
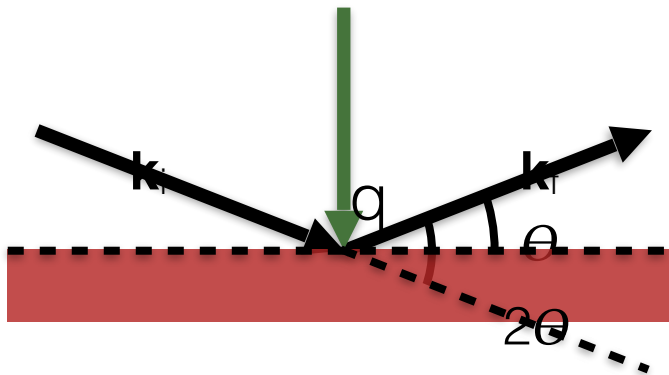




Reflectivity curves

- Reflectivity, super mirror, reflectivity curve

$$m = \frac{\theta_{mirror}}{\theta_{Ni}}$$

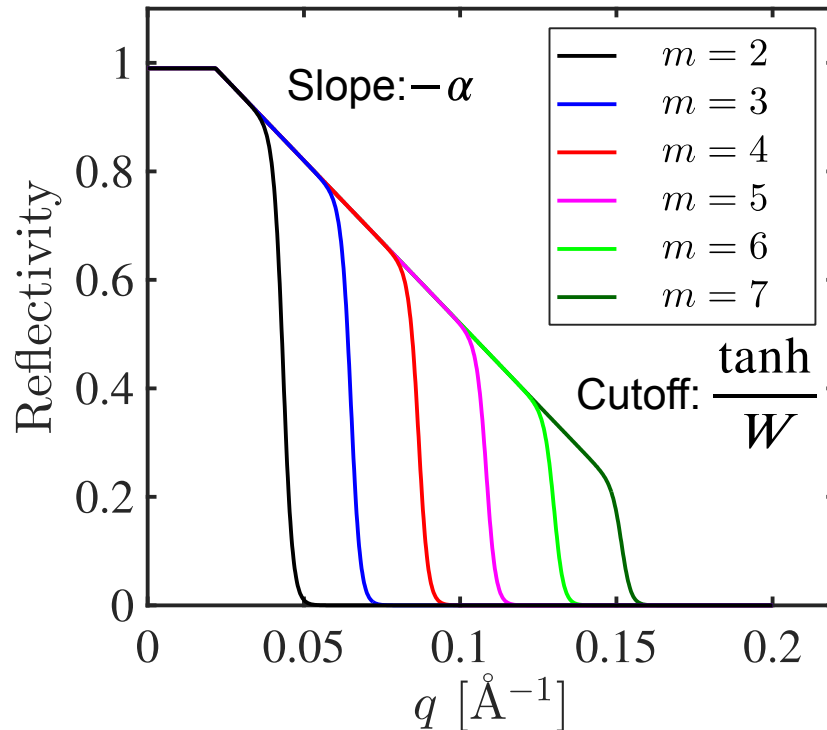




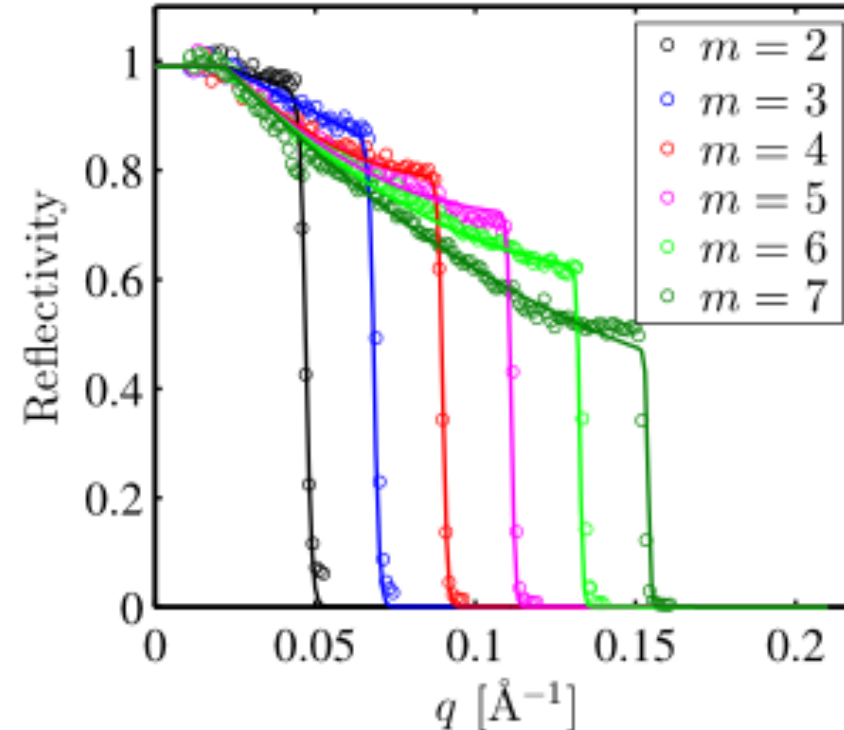
Reflectivity curves in McStas

$$R(q) = \begin{cases} R_0 & \text{if } q < q_c \\ R_0(1 - \tanh((q - mq_c)/W))(1 - \alpha(q - q_c))/2 & \text{otherwise} \end{cases}$$

McStas standard model



McStas fitted model

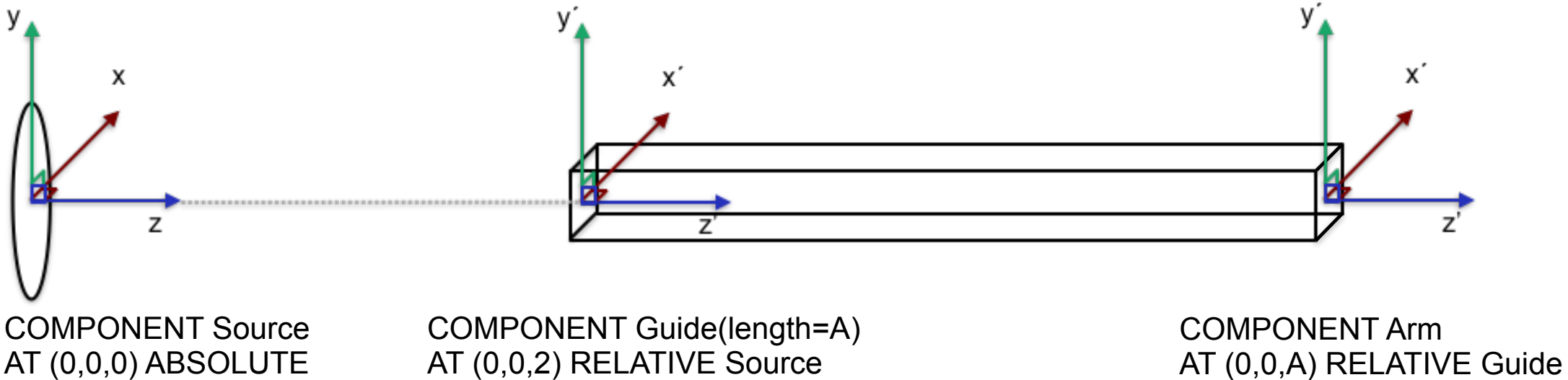


$\alpha = 0$
 $W = 0$
 Only m matters
 Better mirrors available today



Guide placement in McStas

- The center is the front of the guide element
- Tip: Insert a guide at the end of the guide



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Gravitation in McStas

- Enabled by adding `-g` / `--gravitation` on command line or by selecting “Gravity On” in mcgui

- Default ~ gravity on earth

`#define GRAVITY 9.81 /* [m/s^2] gravitational acceleration */`

(If on the moon, use `-DGRAVITY=1.62` ;-))

- For guides, only `Guide_gravity` and `Elliptic_guide_gravity` support parabolic propagation. (Many others propagate linearly in $\vec{v}$ direction.)

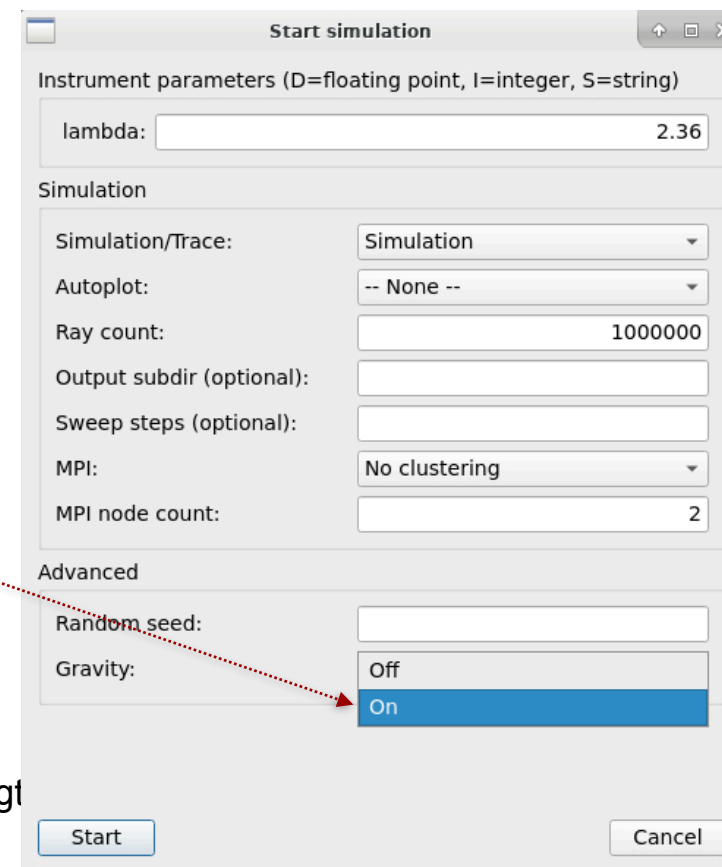
- As you will see in the practical, implications are greatest with long wavelength

- “How about e.g. elliptic mirror optic X that does not support gravity?”

- often a good workaround is to add a monitor close to the surface of object X, this takes care that propagation up to the monitor includes gravitation:

- Gravity is enabled in any call to `PROP_DT`, `PROP_Z0` etc., but not in `intersect_*` routines (most monitors use `PROP_Z0` directly, no `intersect_*` call first)

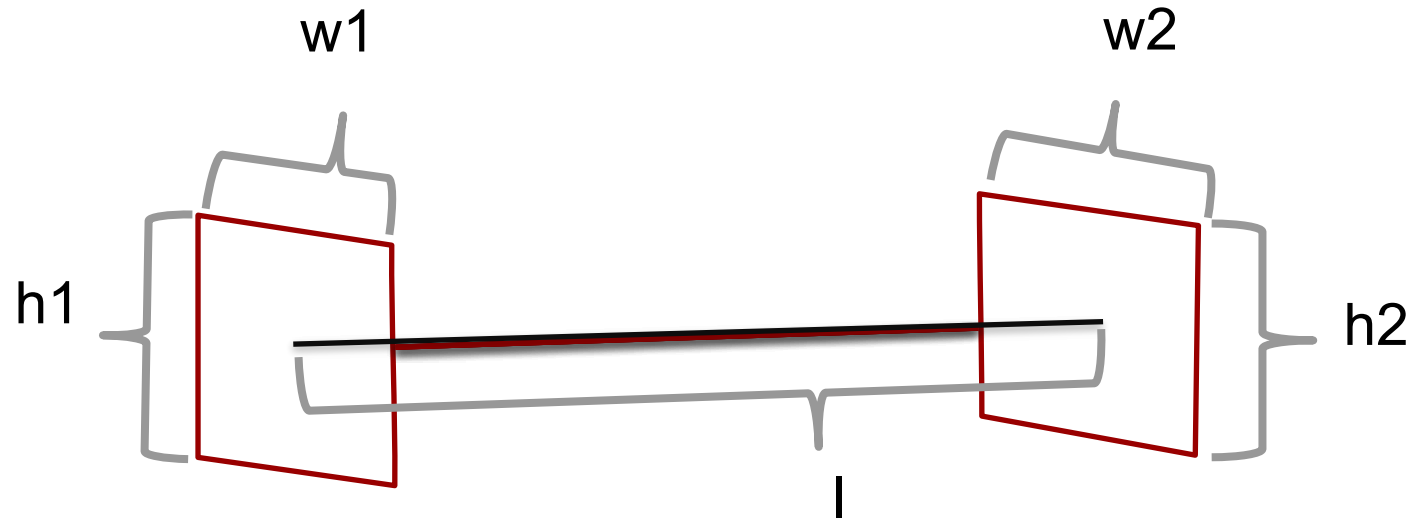
- OK to propagate without gravitation e.g. within sample, through velocity selector etc. / range of ~cm’s





Popular guide components: Guide_gravity

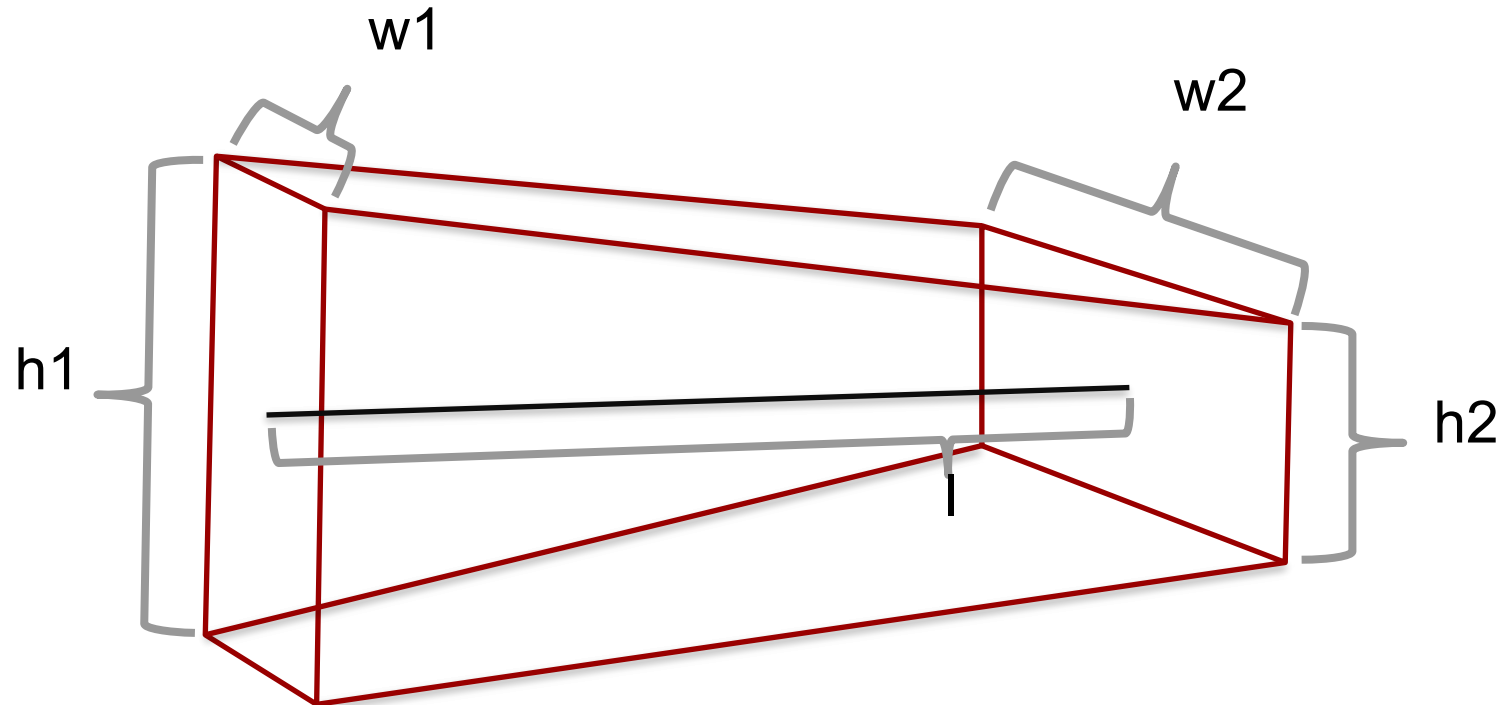
- Typical guide component with gravity, parameter-interface similar to e.g. Guide.comp
- Many additional features, channels, fermi chopper, ... (see mcdoc pages for more info)





Popular guide components: Guide_gravity

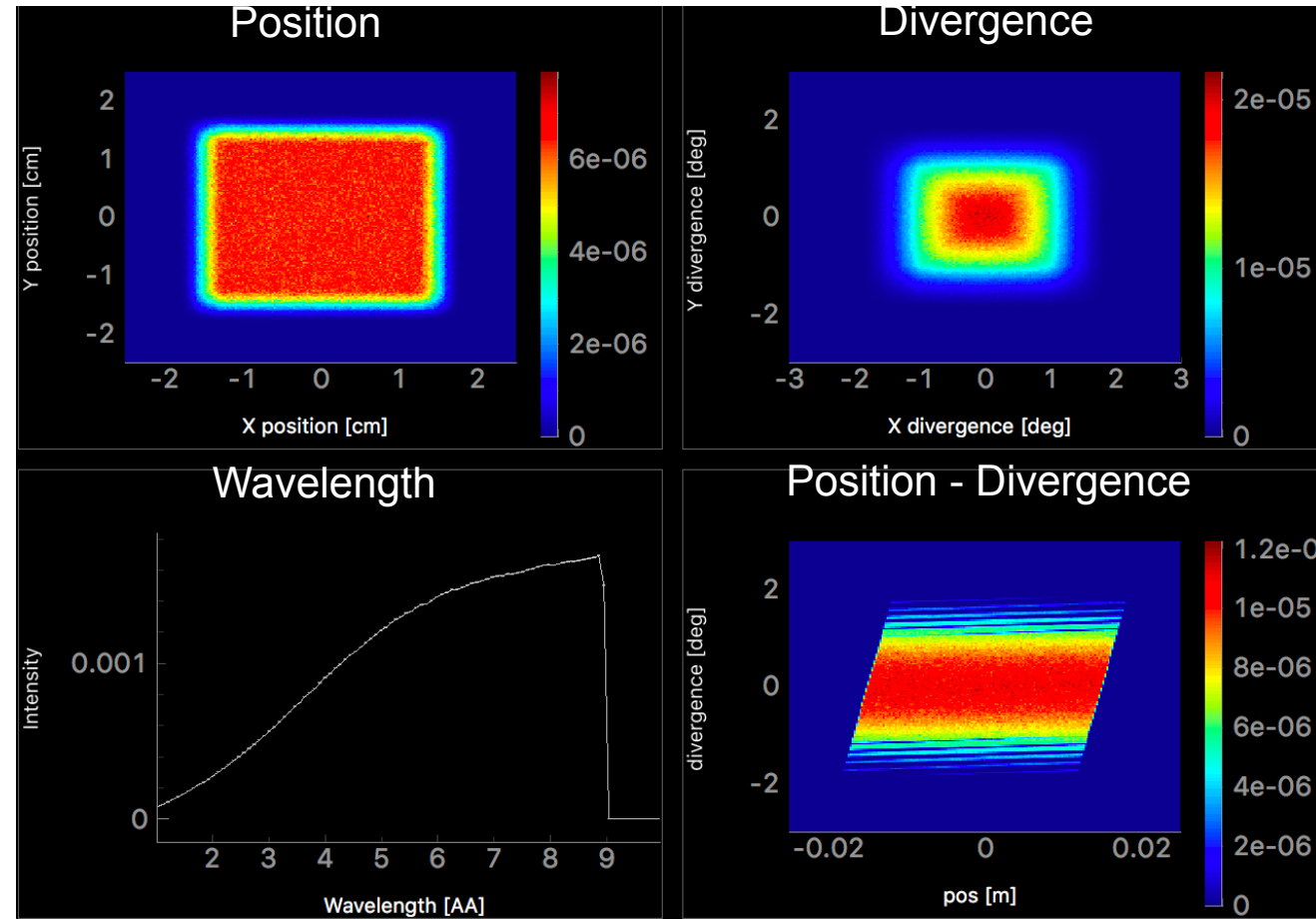
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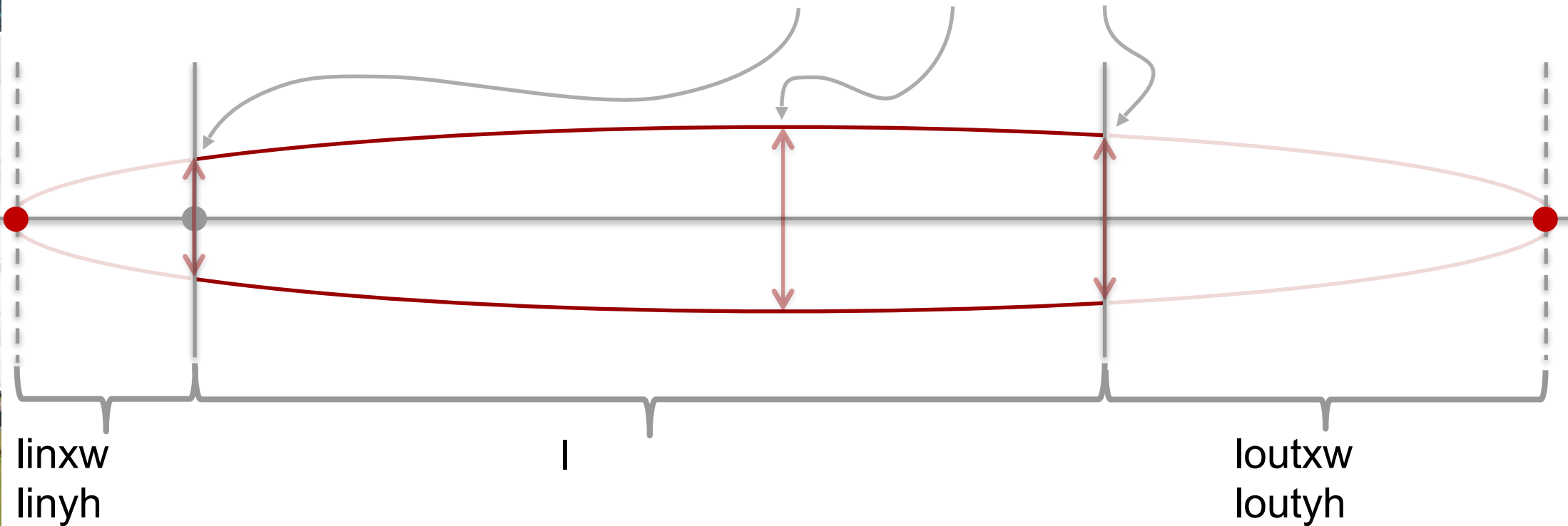
- Typical guide component with gravity



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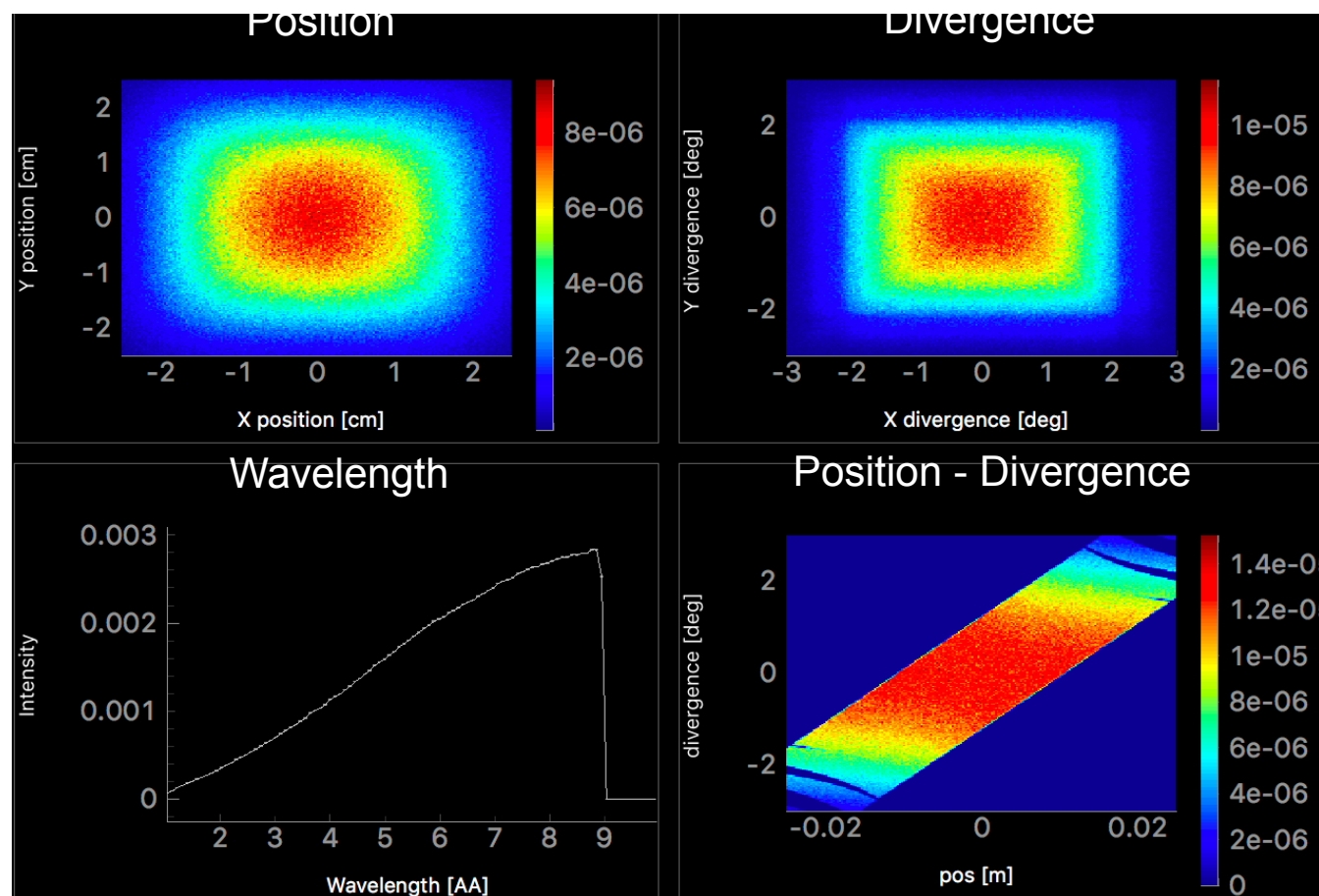
Popular guide components: Elliptical_guide_gravity

- Useful for elliptic and parabolic guide geometries, focusing, ballistic, coating distribution, xwidth and yheight at DimensionsAt = "entrance", "mid" or "exit"



Popular guide components: Elliptical_guide_gravity

- Useful for elliptic and parabolic guide geometries, focusing, ballistic, coating distribution,
...

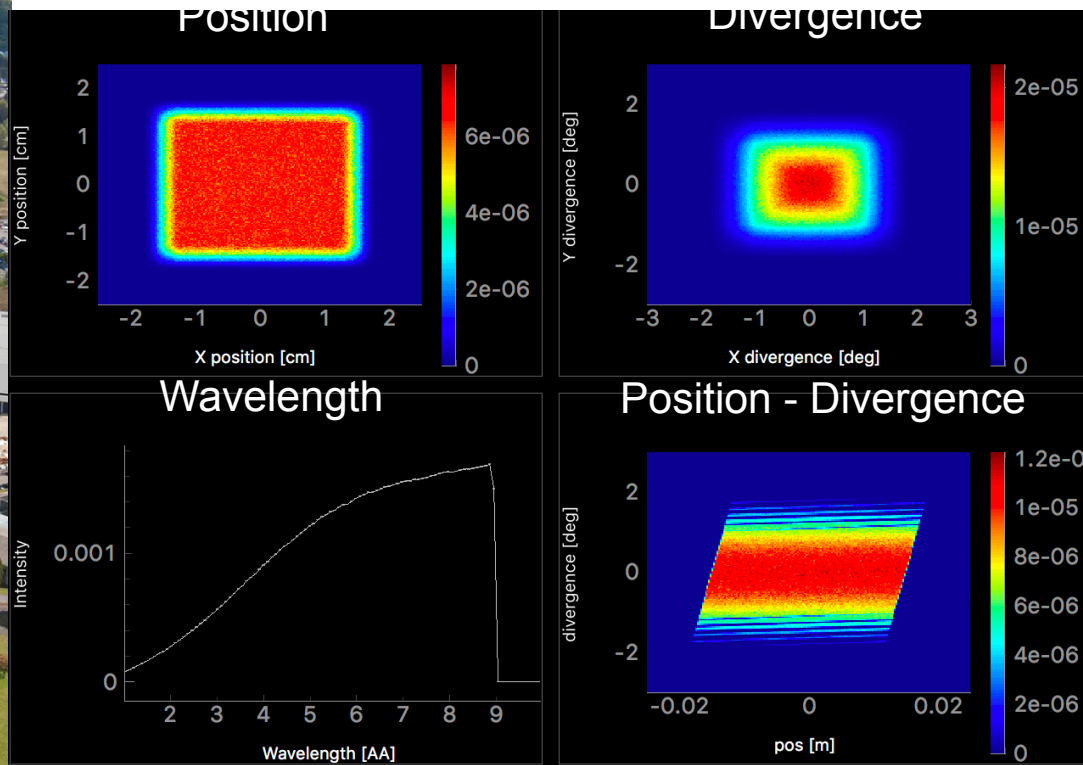


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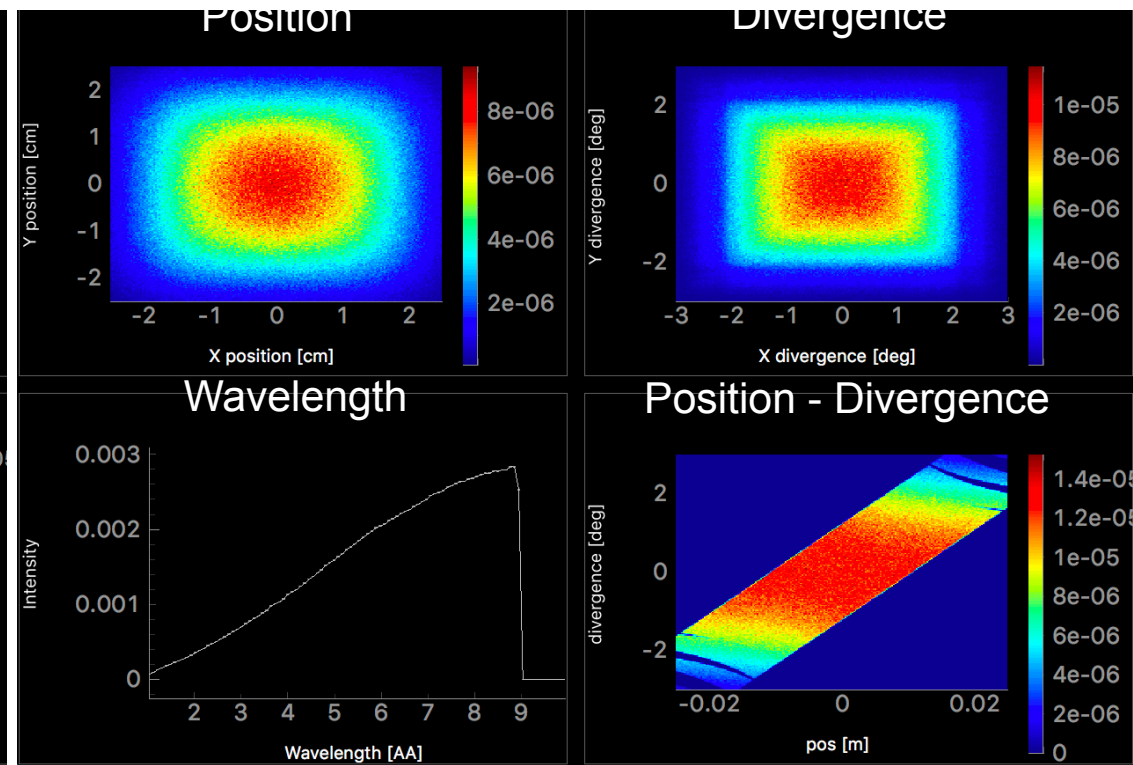
Comparison: Guide_gravity and Elliptic_guide_gravity



Guide_gravity



Elliptic_guide_gravity

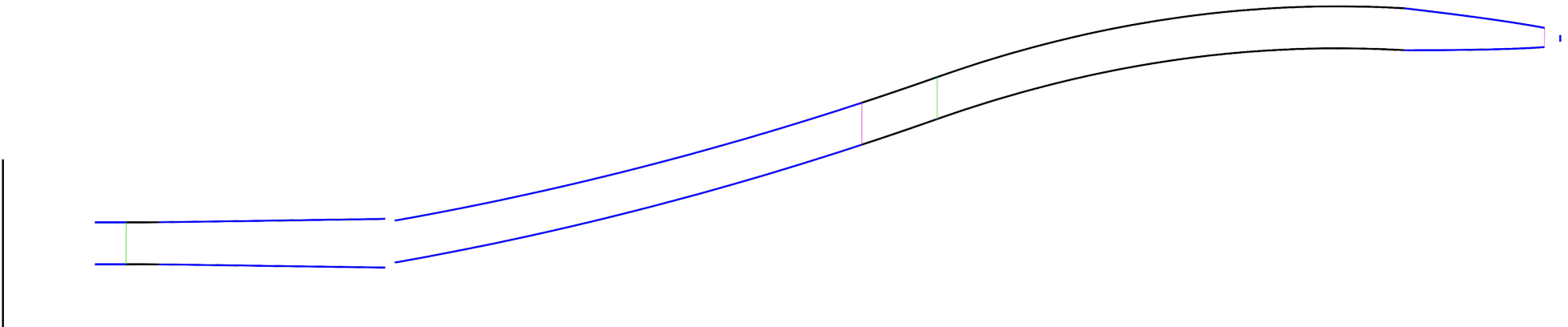


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Breaking line of sight

- Importance of breaking line of sight, ways of doing so, ...

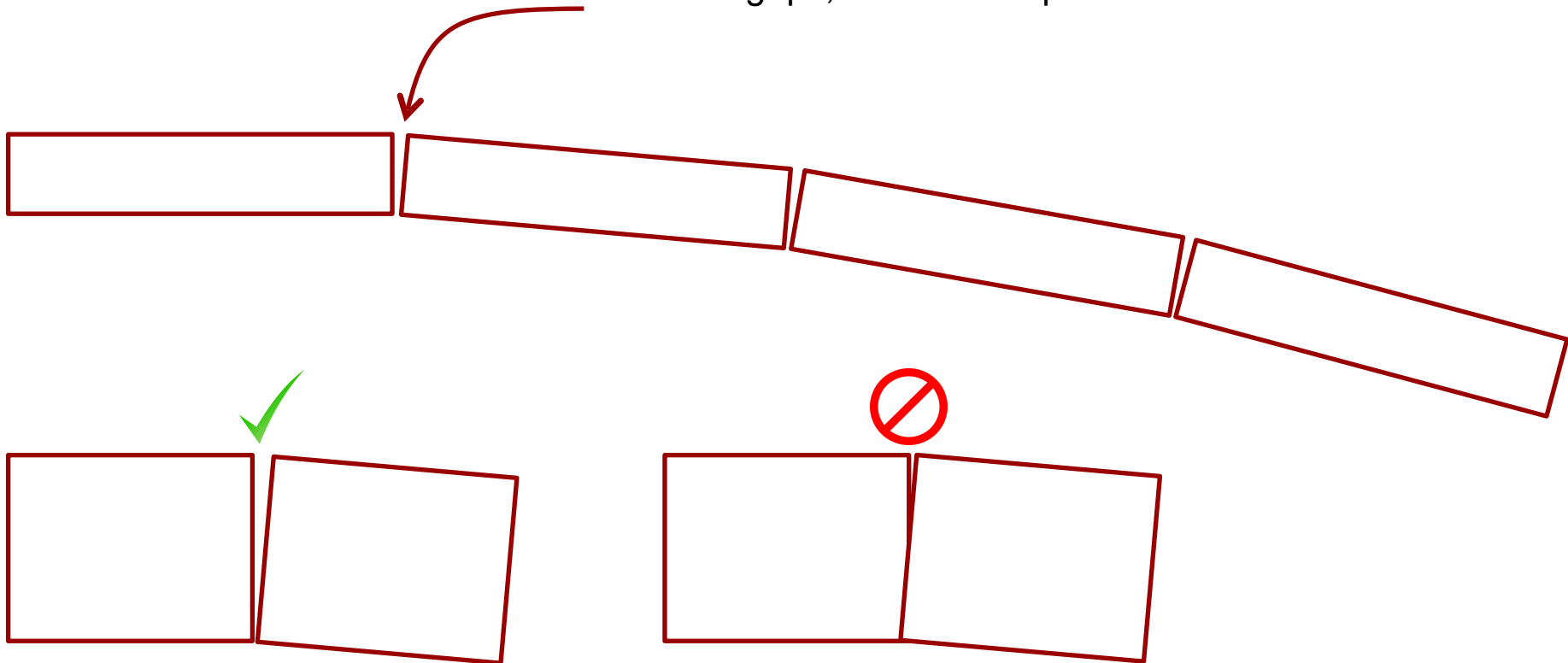


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Breaking line of sight

- Bender / Guide_curved component or many straight sections
Mind the gaps, avoid overlap

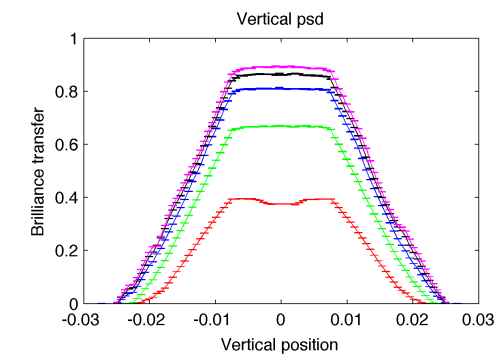
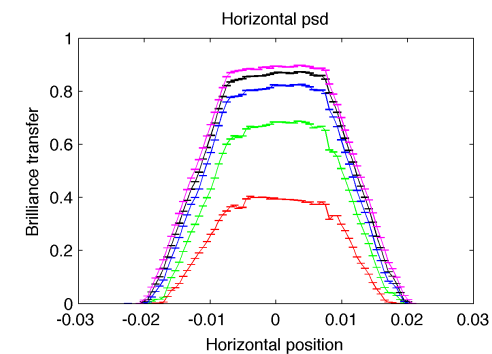
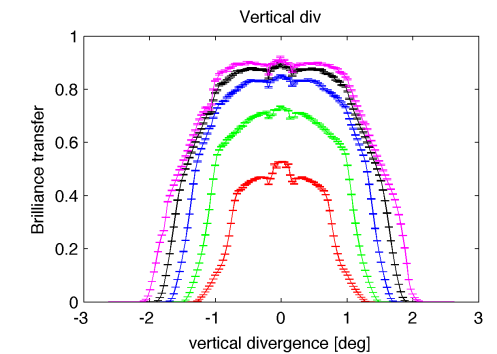
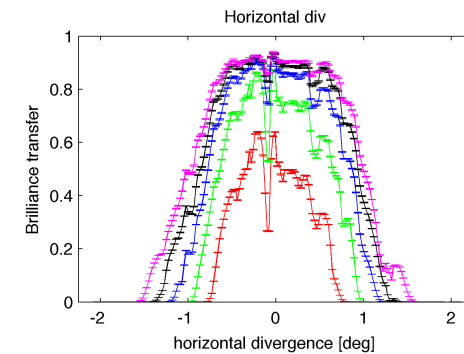
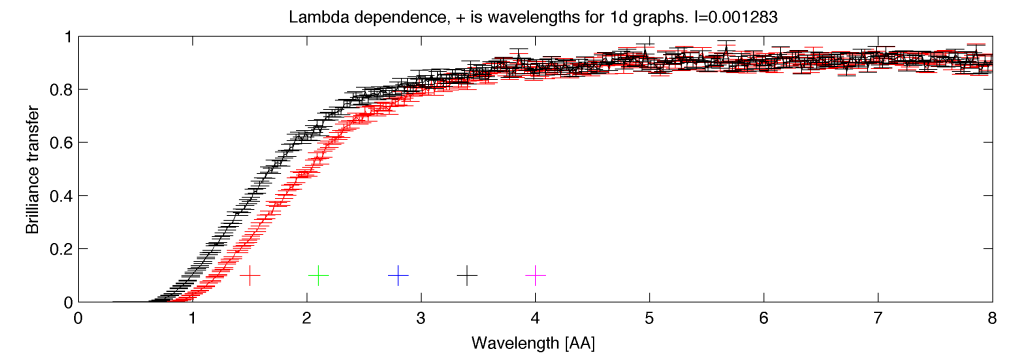
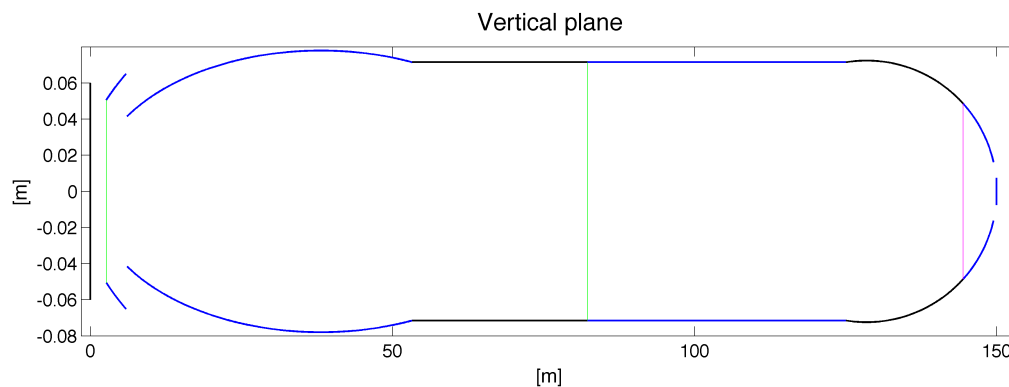
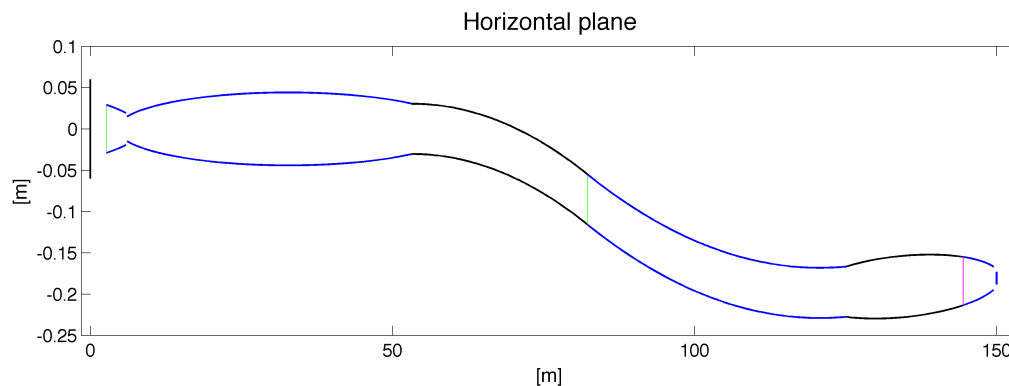


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A guide design

- To be continued in Thursday guidebot talk by Mads Bertelsen



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Guide exercise

- Insert a guide and use an instrument input parameter to set the length
- Use monitors to see the resulting beam
 - PSD_monitor (spatial distribution)
 - Divergence_monitor (divergence distributoin)
 - L_monitor (wavelength distribution)
 - Posdiv_monitor (acceptance diagram)
- Extra tasks:
 - Scan guide length
 - Introduce a gap by using two guide components
 - Use Guide_gravity and extend to 100 m length
 - Investigate the effect of gravity on the transport of long-wavelength neutrons